Endogenous public amenities*

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Abstract

This paper examines how local public amenities are distributed across neighbourhoods within cities. Using novel comprehensive data on libraries, parks, pools, leisure centres, and infrastructure investments in England (2005–2024), I document a striking U-shaped relationship between neighbourhood affluence and public amenity access: the poorest and wealthiest neighbourhoods have the best access, while middle-income areas are systematically underserved. To address endogeneity from household sorting and amenity capitalisation, I exploit quasi-experimental variation from the 2014 Stamp Duty Land Tax reform, which generated discontinuous changes in transaction costs across the house price distribution. Instrumental variable estimates confirm the U-shaped pattern. Investigating mechanisms, I show that poor neighbourhoods exhibit high demand for public amenities, while rich neighbourhoods have greater political power. In contrast, private amenities like restaurants or bars concentrate exclusively in affluent areas. A quantitative spatial model with endogenous public amenity provision rationalises these patterns through political economy: local governments balance redistribution toward needy neighbourhoods and capture by politically powerful wealthy neighbourhoods, leaving the middle class behind. The findings reveal a novel form of spatial inequality within cities and have implications for local public finance, urban policy, and quantitative spatial models.

Keywords: Public amenities, local public goods, political economy, neighbourhood inequality, quantitative spatial models, urban economics
JEL Codes: D72, H41, H72, R23, R31, R51

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1 Introduction

Local public amenities such as libraries, parks, and leisure centres shape where people live (Roback, 1982), how children grow up, and which neighbourhoods thrive (Chetty and Hendren, 2018). A large body of research documents that households value proximity to public amenities and are willing to pay substantial premiums to live near parks, libraries, hospitals, and schools (Bayer et al., 2007; Brueckner et al., 1999; Castanheira et al., 2025; Gibbons and Machin, 2003; Gibbons et al., 2013; Gilpin et al., 2024; Hanzl and Gilpin, 2025; Karger, 2021; Koster, 2023; Minano-Manero, 2025; Pietrabissa, 2024; Puente-Beccar, 2025). As a result, unequal distribution of public amenities across space manifests in unequal access across income. Yet public amenities are not exogenous features of cities: they are created and allocated by city officials and local politicians. This paper studies how neighbourhood composition affects the allocation of public amenities within cities, and how political incentives can narrow or entrench within-city inequality.

This question asks the converse of the classic sorting problem: not only how amenities shape who lives where, but how the income composition of neighbourhoods shapes which amenities are provided. The answer to that question is theoretically ambiguous. Equity and efficiency concerns, coming from the public character of these amenities, may suggest to use public amenities as a policy tool to promote equity. In a city, this means allocating public amenities in more deprived neighbourhoods to upgrade them through investments in libraries, parks, and community facilities. This suggests a negative relationship. Alternatively, a parallel literature in political economy suggests that politicians allocate public resources following political forces like favouritism, lobbying, or strategic allocation towards swing districts, co-ethnic regions, or politically connected constituencies (Baskaran and Lopes Da Fonseca, 2021; Franck and Rainer, 2012; Grossman and Helpman, 1996; Hodler and Raschky, 2014; Larcinese et al., 2006; Solé-Ollé and Sorribas-Navarro, 2008; Olson, 1965; Weingast et al., 1981). This would result in a positive relationship between public amenity access and neighbourhood affluence. Without empirical evidence, theory alone cannot settle whether amenities are distributed equitably, skewed toward the rich, or directed toward the poor. The findings, thus, have direct implications for urban policy making: if middle- or low-income neighbourhoods systematically receive fewer public amenities than the rich, this could reinforce spatial sorting, deepen neighbourhood stratification, and undermine social cohesion.

To answer this question, I assemble a new dataset that combines stock and flow measures of local public amenities for the whole of England. On the stock side, I collect novel information on the universe of public libraries, public pools, and leisure centres from 2005 to 2024 via Freedom of Information requests to all 317 local authorities, complemented with administrative Points of Interest (POI) data for libraries, parks, museums, community centres, and related amenities. On the flow side, I build a new measure of local infrastructure investments by classifying public procurement tenders into infrastructure versus non-infrastructure contracts using a large language model and geolocating them via postcodes. These data are matched to neighbourhood-level measures of affluence based on the universe of property transactions, as well as measures of education, income estimates, and demographic composition.

I then proceed in three steps. First, I quantify the relationship between neighbourhood affluence and the presence of public amenities. The central empirical result is surprising. Using high-resolution data on roughly 34,000 neighbourhoods in England between 2005 and 2024, I document a robust U-shaped relationship between neighbourhood affluence – measured primarily by average property transaction prices – and access to local public amenities: the poorest and the richest neighbourhoods have the best access, while middle-income neighbourhoods lose out. Poor neighbourhoods tend to be close to public libraries and other amenities, and so are the wealthiest neighbourhoods; it is the "middle" of the house price distribution that is physically furthest away and least well served. This pattern holds across different measures of access (distance to the nearest amenity, presence within the neighbourhood, presence within 1 km) and for an aggregate amenity index constructed from multiple amenity types, as well as for different amenity types.

Establishing this U-shaped pattern is challenging because neighbourhood affluence and amenity provision are jointly determined. Amenities capitalise into house prices, and households sort into neighbourhoods partly based on amenity availability, generating severe endogeneity concerns. To account for that, I exploit quasi-experimental variation from the 2014 reform of the UK Stamp Duty Land Tax (SDLT), which discretely changed transaction tax rates at different price thresholds. By constructing neighbourhood-level exposure measures to the reform and using them as instruments for average transaction prices, the paper isolates plausibly exogenous variation in neighbourhood affluence and shows that the U-shaped relationship persists in instrumental variable specifications.

Second, I explore mechanisms. Using survey data on library usage, I show that children from low-income households use public libraries much more frequently than children from high-income households, consistent with high demand for publicly provided amenities in poor neighbourhoods. At the same time, poor and rich neighbourhoods both receive more public amenities than middle-income neighbourhoods, whereas private amenities such as bars, cafés, and restaurants concentrate almost exclusively in affluent areas. I show that public amenities are disproportionately located in neighbourhoods with high political power and that residents of affluent neighbourhoods submit more complaints about neighbourhood conditions. Taken together, these patterns suggest that public amenity allocation reflects both redistribution toward needy neighbourhoods and capture by politically powerful, affluent neighbourhoods, leaving the urban middle class relatively underserved.

Third, building on the reduced form findings, I develop a quantitative spatial equilibrium model with endogenous public amenity provision. Households with heterogeneous income and preferences choose residential locations based on wages, housing costs, commuting access, and the availability of both public and private amenities. Local politicians allocate public amenities across neighbourhoods by maximising a weighted combination of overall welfare and political power of the two groups. The model nests two competing mechanisms: politicians may favour affluent neighbourhoods due to their disproportionate political influence, or they may target poor neighbourhoods to address social needs and maximise aggregate welfare gains from public investment. I calibrate the model for the city of Birmingham and show that the model reproduces the U-shaped

relationship between public amenity access and neighbourhood affluence across space. Using this calibration, I evaluate a policy counterfactual in which I remove political influence from the policymaker's objective. Public amenities are reallocated towards more deprived neighbourhoods, which increases city-wide welfare and decreases within-city inequality. This shows that the allocation of public facilities can be an important policy tool to reduce spatial inequalities.

Related literature This paper contributes to three strands of literature. First, it adds to the emerging literature on endogenous amenities in spatial models. While quantitative spatial models have traditionally treated amenities as exogenous location fundamentals or modeled them as a distance-weighted function of the mass of people living nearby (Ahlfeldt et al., 2015; Redding and Rossi-Hansberg, 2017), recent work has begun to unpack the determinants of amenity provision and model amenities as endogenous outcomes of economic activity (Ang et al., 2024). Almagro and Domínguez-Iino (2025) and Leonardi and Moretti (2023) develop frameworks where private amenities (restaurants, bars, retail) respond to local demand and income. Bagagli (2025), Bordeu (2025), and Fajgelbaum et al. (2024) incorporate political economy mechanisms such as segregation, coordination, and voting into the provision of public transport infrastructure. My paper extends this literature by focusing explicitly on *local* public amenities—such as libraries, parks, and leisure centres—and modelling their supply through political economy mechanisms at the neighbourhood level rather than market forces. This distinction is important because public amenities are provided by local governments facing political incentives rather than profit-maximising firms, leading to different spatial allocation patterns.

Second, I contribute to research on public goods within cities. Recent studies have examined the spatial distribution of various public goods and services: Asher et al. (2025) study infrastructure provision in Indian cities, showing that schools and hospitals are regressively distributed with respect to caste. Castanheira et al. (2025) analyse hospital and school accessibility in Belgium and show that public goods favour rich neighbourhoods. Similarly, Fabre (2025) confirms this finding for higher education opportunities in France. Harari (2024) examines urban service provision in developing countries and provides evidence that segregation affects provision patterns. My paper adds to this literature by documenting a U-shaped relationship between neighbourhood affluence and public amenity provision and by developing a model with explicit political economy microfoundations for local amenity allocation decisions.

Third, the paper intersects with the political economy literature on local public goods provision. A large body of work shows that political incentives—including favouritism towards coethnics (Franck and Rainer, 2012; Hodler and Raschky, 2014), pork barrel spending in swing districts (Larcinese et al., 2006; Solé-Ollé and Sorribas-Navarro, 2008), and the influence of special interest groups (Grossman and Helpman, 1996; Olson, 1965; Weingast et al., 1981)—shape the spatial allocation of public resources at regional and national scales. Baskaran and Lopes Da Fonseca (2021) provide comprehensive evidence on political targeting in public employment. My contribution is to bring these political economy insights into urban economics by documenting how political power operates within cities at the neighbourhood level. By showing that public

amenities concentrate in both poor (high-need) and rich (high-political-power) neighbourhoods while middle-income areas lose out, the paper reveals a novel form of spatial inequality driven by the intersection of redistributive motives and political capture.

The rest of the paper is organized as follows. Section 2 describes the institutional context of local government finance and political organization in England. Section 3 introduces the data and the construction of amenity and affluence measures. Section 4 presents the empirical analysis of amenity presence and quality, including the instrumental variable strategy based on the SDLT reform. Section 5 discusses mechanisms related to demand and political economy. Section 6 outlines a quantitative urban model that embeds the empirical patterns in a spatial equilibrium framework. Section 9 concludes.

2 Context

I examine the effect of neighborhood affluence on public amenity availability in England between 2005 and 2024. My focus is on amenities provided by local government rather than national government. Large-scale infrastructure projects are not done as frequently and respond less to the local needs of neighbourhoods. Instead, I focus on local government decisions about where to locate libraries, leisure centres, and parks within cities, where the spatial allocation of resources can reflect neighborhood characteristics.

England comprises 317 Local Authority Districts (LADs) of various types, including county councils, district councils, London boroughs, metropolitan boroughs, and unitary authorities. These Local Authorities are responsible for providing public services including waste collection, social care, fire and police services, education, highways, and the provision of public amenities such as public libraries, swimming pools and leisure centres, transport infrastructure, and parks[1]. To understand why England is an ideal setting for this research, it is important to examine how councils allocate their resources.

Local Authority spending Understanding the composition of Local Authority spending is essential to comprehend why England provides an excellent setting for studying public amenity allocation. The largest portion of council spending is on statutory services like social care and fire and police services, with around three-quarters of overall spending. In contrast, cultural amenities—including libraries, museums, galleries, theatres, community centres, leisure centres, and parks account for only 5% of total council spending. This stark difference reflects the distinction between statutory and discretionary services. While councils face legal obligations to provide social care, education, and emergency services, most cultural amenities are provided on a discretionary basis. Although libraries are technically mandatory under the Public Libraries and Museums Act 1964, councils retain substantial discretion in determining what constitutes 'comprehensive and efficient' service provision. This discretion extends to most other cultural amenities, which are

[1]Unitary authorities are responsible for all services while in two-tier councils, the county and district councils divide the services between them. In my empirical analysis, I control for the relevant authority and cluster standard errors at that level.

provided on a purely discretionary basis.

This institutional structure creates a fiscal environment where public amenities serve as the primary margin of adjustment when Local Authorities face financial pressure. Between 2010-11 and 2023-24, while total service expenditure increased in nominal terms, spending on cultural services declined by £472 million (30%) as councils protected statutory services amid rising demand and constrained revenues. The discretionary nature of most public amenities, combined with growing statutory obligations, makes England an ideal setting for examining how neighborhood-level political economy factors influence the spatial allocation of scarce public resources.

Local Authority revenue Local government in England operates within a centralized fiscal system with limited local revenue autonomy. National government grants comprise approximately 70% of Local Authority revenue, making them the largest revenue source. The remaining portion comes from retention of business taxes, local fees, and council taxes. Council tax, the primary source of locally-controlled revenue, is levied annually on residential properties based on their 1991 property values. Properties are assigned to one of eight bands (A through H) based on these historical valuations. Critically for this analysis, property valuations for tax purposes remain frozen at 1991 levels. Increases or decreases in property values due to housing market changes do not alter the council tax band, meaning neighborhoods that have gentrified don't experience increases in their local tax base, nor do deteriorating neighborhoods see tax reductions. This is important for my analysis as this makes it unlikely that councils locate amenities in response to the council tax revenue of some neighborhoods.

Political system The spatial allocation of discretionary resources reflects both fiscal constraints and political incentives. The political system governing Local Authorities is structured as follows: Local Authorities in England are governed by elected councils operating under either a Leader and Cabinet model or a committee system. Under the Leader and Cabinet model—adopted by most councils between 2000 and 2011—the political party or coalition with the most seats forms a cabinet that makes executive decisions, with a council leader elected by fellow councillors. The Localism Act 2011 gave councils the option to return to committee systems, and some have done so. 27 councils have directly elected mayors who perform the leader's role but are elected by residents rather than councillors.

3 Data

Unit of observation Before I start outlining the different data sources, it is important to clarify the unit of observation that I use in my analyses. As I am interested in differences in accessibility of public amenities within cities, my unit of observation will be neighbourhoods. I define lower layer super output areas (LSOAs) as a neighbourhood. This is a statistical unit in the UK aggregated from census blocks (also called output areas) and constructed to have, on average, a comparable size in terms of population. LSOAs have an average population of 1500 people or

650 households and are thus comparable to US census block groups. Within cities, they can be understood as neighbourhoods, while in rural parts, the area is significantly larger. I will control in all my analyses for urban versus rural neighbourhoods and for neighbourhood population density. Throughout this paper, I will refer to these units as "neighbourhoods". My sample comprises of roughly 34,000 LSOAs.

3.1 Data on public amenities

To analyze the effect of neighbourhood affluence on the availability of public amenities, I build a novel and comprehensive dataset on public amenity accessibility for England. The dataset covers the period from 2005 to 2024 and includes both stock measures (the presence of physical amenities) and flow measures (investments in amenity quality and local infrastructure). The reason to include both the stock and the flow of local public goods is that it usually takes a fair amount of time to build, for instance, a new library. A flow measure of money being directed towards a library gives information on the quality improvements of a physical stock of a given public amenity. In the following two paragraphs, I will explain these two kinds of public amenities.

Stock of public amenities The information on the physical stock of public amenities comes from two main sources. First, I collect novel data on public libraries, public pools, and public leisure centres in England by sending out Freedom of Information (FOI) requests to all English local authorities. In these requests, I ask for the following information for the period 2005 – 2024: For each facility (library, pool, or leisure centre) that was open during that period, I request the current or historical address, year of opening, year of closing (if applicable), and average weekly opening hours in each year. These amenities do not represent the largest share of local authority spending which goes traditionally to spending on social care. However, spending on social care is statutory leaving little room to adjust and change spending there. The amenities I study, on the other hand, are cultural amenities that make an area nice to live in, and over which local authorities have much more discretion when it comes to the financing.

In the United Kingdom (UK), FOI requests are a powerful tool to ask for recorded information held by public bodies. Under the Freedom of Information Act of 2000, these bodies are obliged to reply to the request. They can refuse the request on different grounds, among others, if the information asked for is publicly available, if the information is classified, or if providing the information asked for would result in unreasonably high costs to the public body asked. The FOI requests resulted in an 80% success rate of retrieving answers and collecting data from local authorities (see Appendix Figure A1). I supplement the information on libraries with data from the Chartered Institute of Public Finance and Accountancy (CIPFA), an association of public accountants in the UK that provides reports and data on local government finances and services. For the years 2016, 2021 – 2024 they provide official data on each public library in the UK with the same information that I collected and additional variables such as the staffed hours. This allows me to validate the FOI data and obtain information from authorities that did not respond to FOI requests.

The second source is the Points of Interest (POI) data provided by Ordnance Survey. This dataset contains all public and private businesses, as well as education and leisure services. I have access to the data for the years 2014 – 2024. I filter the dataset to get information on public libraries, parks, museums, community centres, Wi-Fi hotspots, and public toilets, as well as public "bads" such as recycling centres. This administrative dataset provides me with the universe of points of interest in the UK. This means that for all available years I know the stock of all public amenities mentioned above. The POI data span only 2014–2024, limiting longitudinal analysis. In particular, this period does not align with the 2011 and 2021 census years, which provide the only years with detailed demographic data at the neighbourhood level. Furthermore, in these variables, I only know the extensive margin of public amenity presence. The data from the FOI requests enhances this information by providing intensive margin measures via the opening hours.

Flow of public amenities Both the POI data and the data from the FOI requests give me extensive margin measures of the physical presence of a public amenity like a library or a pool. Building a new library or pool is a large-scale programme that can take a long time and comes with several bureaucratic hurdles, like getting approval for your planning application. Thus, there is very likely not much variation over time in the accessibility of these amenities. To overcome this issue, I rely on a measure of infrastructure investments using public procurement tender data from the EU's TED database (2009–2024). The data are at the procurement tender level and provide information on the tender, like title, description, type, or price, as well as buyer and supplier information. Each tender represents a contract that local authorities have announced. I first filter the data on whether the buyer of the tender is a local authority, and then on whether there is a postcode of implementation available. Postcodes of implementation are not available for every tender but only for a subsample of 70%. Tenders don't have a postcode of implementation if there is no clear location identifiable. This is the case, for instance, if a local council hires a new officer or if they are buying new vehicles for their waste disposal department. This leaves me with a subset of procurement tenders for the UK of 5731 tenders. Not all tenders are infrastructure investments; many involve service contracts, vehicle purchases, or administrative hires. While there is a broad categorization of tenders into Supplies, Services, and Works provided in the data, I use a local large language model to classify the tenders into infrastructure investments and other tenders. The prompt I run is the following, where the text is composed of the title and the description of the tender. I provide an example of a tender text that was classified in appendix section C.

> *"I am giving you the extracted title and description from a UK procurement tender. The buyer of the tender is a UK local authority. Your task is to classify whether the tender represents an investment into local infrastructure.*
>
> *Local infrastructure investments can either be physical investment in the form of repairs, newly built infrastructure or investments into services that likely affect the quality of public goods or public services.*
>
> *Please classify the tender into whether it is an infrastructure investment or not. If it is an investment into local infrastructure, extract the type of investment. Lastly, please flag if this*

is an investment into a library.

Please return the output in structured JSON format with three keys: d_investment, investment_type, d_library.

Here is the document text:

[Followed by title and description]

I refine the prompt in an iterative manner, giving examples of misclassified tenders to avoid errors. The resulting subset of classified infrastructure investments is 5298 tenders. These include investments like the repair of a library's roof or the building of a new playground in a park. I manually cross-validate a 10% sample of the classified tenders.

To give more context on the procurement data, Panel A of Figure 1 shows the number of tenders in the data for which I know the postcode of implementation by year and type of tender. The tender types are given by the data as explained above. There is variation over time in the number of tenders, with up to 3000 tenders in a given year. Panel B of the figure shows how these types translate into infrastructure investments as classified by the LLM. The figure shows that among tenders labeled as Works, virtually all are classified as infrastructure investments. For Supplies and Services, the share classified as infrastructure investments is lower but still high, with around 70%.

Access to public amenities For each neighbourhood in my dataset, I construct different measures of access to each public amenity and infrastructure investment. I first create a population-weighted centroid of each neighbourhood and then create three types of access measures: distance of each centroid to the closest amenity/infrastructure investment, presence of an amenity/infrastructure investment within a neighbourhood, and presence of an amenity/infrastructure investment within a radius of 1 km of the centroid. I subdivide the last two measures into a discrete and continuous variable. To provide some information on these measures, Table 1 shows summary statistics for each type of access measure and each type of amenity.

Creation of a public amenity index Different amenities are likely distributed in different ways within a city. In addition, I want to provide an average figure of amenity access for neighbourhoods. To that end, I create an amenity index in two ways. The first way is to standardize each access measure for each type of public amenity and average them, i.e., given for each measure of access the K amenities $X_1, X_2, ..., X_K$, I compute the standardized values of the access measure to each amenity $\hat{X}_{itk}$, like standardized distance to the closest public library in year t. The simple average amenity index is then:

$$\text{AmenityIndex}_{it} = \frac{1}{K} \sum_{k=1}^{K} \hat{X}_{itk}$$

where K is the number of amenities. This index represents the mean standardized accessibility across all amenity types. This results in five averages of amenity access, one for each type of access measure.

The second way follows Cook et al. (2023) and Diamond (2016), who use the first principal component of a principal component analysis (PCA) and interpret it as an amenity index. This first component is defined as

$$\text{PC1}_{it} = \sum_{k=1}^{K} w_k \hat{X}_{itk}$$

where $\hat{X}_{ikt}$ are standardized values of a given access measure of neighbourhood i to amenity k in year t. w_k are loadings of this variable on the first primary component, which satisfy $\sum_k w_k^2 = 1$ and are chosen to maximise the variance among all possible weighted averages.

Again, I do this for every type of access measure, which results in five amenity indices. The simple average index assumes all individuals value amenities equally, a limitation recognised in the literature on heterogeneous amenity preferences (Ahlfeldt et al., 2025; Albouy and Faberman, 2025; Almagro and Domínguez-Iino, 2025; Cook, 2024; Couture et al., 2024; Deffebach et al., 2025; Diamond and Suárez Serrato, 2025). The PCA index identifies a common latent factor driving joint amenity distributions without imposing explicit preference weights. Appendix Figure A2 shows that these two indices are highly correlated, suggesting they capture the same underlying amenity access variation despite different conceptual approaches. While most literature uses PCA indices as explanatory variables, I use the first principal component as a dependent variable to test whether neighbourhood affluence predicts overall amenity access. Although the PCA coefficients lack direct economic interpretation (representing a latent amenity index rather than a specific amenity type), this approach provides a parsimonious test of the overall amenity-affluence relationship before examining specific amenity types.

Additionally, I leverage the POI data to get information on private amenities, specifically bars, cafés, and restaurants, to contrast them with the findings for public amenities. I create the same amenity indices for private amenities as for public amenities.

Table 1 presents summary statistics on public amenity access: The average neighbourhood observes 0.35 libraries and 0.28 infrastructure investments over the period of observation. Interestingly, there is significant variation in the opening hours of public libraries in my data, hinting at the fact that even if a neighbourhood has access to a public library, the quality of that library might significantly differ from that of other libraries.

3.2 Measures of neighbourhood affluence

I use the term "neighbourhood affluence" descriptively to refer to variables measuring a neighbourhood's economic well-being, without normative implications.

Transaction prices As the main measure, I use average property transaction price from the HM Land Registry, which provides comprehensive transaction data and has become standard in UK spatial economics research (Ahlfeldt et al., 2024; Kleven et al., 2013; Koster, 2023). The data is at the address level, covers the period 2005 – 2022, and records the price for which the property was sold. To bring the data to my unit of analysis, a neighbourhood, I calculate the

average transaction price of all transactions that happened in a given year and neighbourhood. The benefit of using this measure as the main one in my analysis is the granularity and coverage of the data. Furthermore, property transaction prices should capture neighbourhood deprivation and affluence quite well, being correlated with other measures such as wealth or income. The obvious drawback of using property transaction prices as the independent variable in a regression with access to amenities on the left-hand side is that amenities capitalize into house prices because households sort based on them. This introduces an endogeneity issue in my regressions. I will use an instrumental variable strategy to overcome this issue, leveraging a national transaction tax reform, which I will discuss in further detail in section 4.

Further measures of neighbourhood affluence An alternative measure of neighbourhood socioeconomic status is the share of residents with a university degree from the UK census. However, census data are available only for 2011 and 2021, limiting analysis of changes over time to only one long difference. The last measure to capture how well off or deprived a neighbourhood is, is estimated household income from the small area income estimates provided by the ONS. I use as the main variable from this dataset net weekly household income. Household income is not directly observed at the neighbourhood level. Instead, the Office for National Statistics estimates income using correlated variables such as benefit receipt and council tax band. These estimates are available only at the coarser MSOA level (rather than the LSOA level used in this analysis), limiting their utility for within-city comparisons.

Table 1 shows summary statistics for the measures of neighbourhood affluence described: on average, house prices are £292,000, which is in line with the micro-geographic house price index from Ahlfeldt et al. (2023). The average share of people with a university degree is 30%, and the mean net weekly household income is £631, both of which are in line with official statistics.

3.3 Other data

Age composition (share age 65+) and homeownership rates serve as proxies for political participation, based on evidence that older voters and homeowners are more politically engaged (Ahlfeldt and Maennig, 2015; Ahlfeldt et al., 2017; Barilari et al., 2025). These are detailed below in Section 5 on mechanisms. Additionally, Distance to central business district (CBD)[2], distance to town halls, and buildings' age control for spatial and historical heterogeneity within local authorities. I include an urban/rural indicator to account for differences in amenity provision between city and rural areas.

[2]Measured as the coordinates on Google Maps when googling a local authority, except for the case of London, where I define the centroid of the City of London as the CBD.

4 Empirical analysis

4.1 Amenity presence

I begin the empirical analysis by examining the relationship between neighborhood affluence and public amenity access. The data reveal a striking U-shaped pattern: both the poorest and wealthiest neighborhoods have significantly better amenity access than middle-income neighborhoods. Panels A and B of Figure 2 illustrate this relationship. Panel A shows average distance to the closest library by house price decile, with the first decile representing the poorest neighborhoods, the fifth the median, and the tenth the wealthiest. The shortest distances occur at the extremes (deciles 1 and 10), while the longest distances are in the median decile (decile 5). Panel B demonstrates that this U-shaped pattern extends beyond libraries to the overall amenity index, suggesting it is a general feature of public amenity allocation rather than specific to one amenity type.

Panels C and D show the same relationship in binned scatter plots, controlling for local authority fixed effects and neighborhood characteristics (distance to CBD, distance to town halls, population density, age of properties, and urban–rural location). Even after accounting for these characteristics, the U-shaped relationship persists.

To test this relationship more formally, I run the following OLS regressions

$$y_{ilt} = \delta_{lt} + \beta_1 \log(\text{price}_{ilt}) + \mathbf{X}'_{ilt}\theta + \varepsilon_{ilt} \tag{1}$$

$$y_{ilt} = \delta_{lt} + \beta_1 \log(\text{price}_{ilt}) + \beta_2 \log(\text{price}_{ilt})^2 + \mathbf{X}'_{ilt}\theta + \eta_{ilt} \tag{2}$$

where y_{ilt} measures the amenity access of neighborhood i, located in local authority l in year t. I run different regressions across the access measures and the types of amenities, as well as the amenity indices. Price_{ilt} is the average transaction price of properties sold in neighborhood i, located in local authority l, in year t – my variable of interest. In equation 2, I add a square term of the neighborhood affluence measure to capture the inverted U-shape seen in the raw data. $\mathbf{X}_{ilt}$ is a vector of controls including population density, distance to CBD, distance to town halls, property age, and an urban dummy. Additionally, I control for LAD by year fixed effects (δ_{lt}) so that I isolate purely cross-sectional variation in the relationship of public amenity access and neighborhood affluence. ε_{ilt} and η_{ilt} represent idiosyncratic errors. I cluster standard errors at the local authority district level, which is the unit supplying public amenities in England.

Table 2 presents the results for public libraries as an example of public amenities, and Table 3 presents the results for the amenity index. In each table, the access measure varies across columns, where in columns (1) and (2) it is distance to the closest, in columns (3) and (4) the number of amenities within 1 km, and in columns (5) and (6) the number of amenities within a given neighborhood. I present results with and without controls. Panel A shows the linear case from equation 1 and Panel B adds the square term from equation 2. Looking at public libraries first, the linear relationship between public amenity access and neighborhood affluence seems to be

negative. However, when adding the square term in Panel B, it becomes evident that the square relationship is statistically significant and goes in the same direction as the raw data suggested: with increasing property prices, access first decreases and then increases at some point again.[3]

This pattern holds not only for public libraries as an illustration of a public amenity but also appears when looking at the public amenity index. Again, the linear case suggests a negative relationship between a neighborhood's public amenity accessibility and its affluence. As before, when adding the square term of neighborhood affluence, this relationship seems to be quadratic now instead.

These regression results confirm the visual pattern from Figure 2: the relationship between neighborhood affluence and public amenity access is U-shaped, with the poorest and wealthiest neighborhoods enjoying significantly better access than middle-income areas.

While these regressions provide a more robust view of the relationship between public amenity access and neighborhood presence, the OLS estimate may suffer from endogeneity issues. For instance, we know that people like to live close to nice amenities. If people sort based on public amenities, this will drive up demand for housing in areas close to them, resulting in a reverse causality issue. Similarly, there might be other unobserved factors determining the distribution of public amenities as well as the distribution of neighborhood affluence within cities, resulting in an omitted variable bias. Lastly, as I use property transaction prices to proxy for wealth in a neighborhood, it might well be the case that there is measurement error in the independent variable, which again might lead to biased estimates. To overcome this issue, I employ an instrumental variable strategy to isolate exogenous variation in average property transaction prices in a given neighborhood-year pair.

Instrumental variable strategy To isolate exogenous variation in neighborhood property transaction prices, I rely on an instrumental variable strategy. I leverage the national reform of the Stamp Duty Land Tax (SDLT) in the UK, implemented in December 2014. The SDLT is a transaction tax that buyers of residential property pay. The reform restructured how residential property taxes were calculated, where the previous slab system was replaced with a slice system. Under the previous slab system, a single tax rate was applied to the entire transaction price. The specific tax rate was a function of the property price and applied once a given threshold was exceeded. There were different brackets with increasing tax rates. The reform introduced a progressive slice system, comparable with an income tax system, where different portions of the property price are taxed at increasing marginal rates. Additionally, the policy introduced new thresholds and tax rates.

To give a concrete example, in the previous system (slab), a £150,000 property was entirely taxed at 1%, totaling £1,500, while a £250,000 property was entirely taxed at 3%, totaling £7,500. Under the new system (slice), a £150,000 property is taxed as: first £125,000 at 0% + remaining £25,000 at 2%, totaling £500. A £250,000 property is: first £125,000 at 0% + next £125,000 at 2% + final £0 at 5%, totaling £2,500.

[3] Appendix Tables A1 – A4 show the same regressions but for different measures of neighborhood affluence and confirm this relationship

This creates marginal tax rate discontinuities shown in Figure 3. Around the new thresholds, the policy introduced discontinuous jumps in the marginal tax rates before vs. after the policy, which I will exploit in my analysis.

Specifically, as the changes in tax rates are at the property level, I need to aggregate the policy-induced changes to the neighborhood level. I do this in two different ways and, as such, build two different instruments, which are highly correlated. The first instrument captures the average intensity of the tax policy change in a neighborhood and is defined as $Z_i = \frac{1}{N_i} \sum_{j \in i} \Delta\tau\left(p_j\right)$ where N_i are the number of properties in neighborhood i and $\Delta\tau(p_j)$ is the band-specific change in the property tax rate for each property. The second instrument measures the exposure share and is basically a form of a shift-share instrument where I take the share of transactions in neighborhood i in band b s_{ib} and multiply it with the jump in the marginal tax rate $\Delta\tau_b(p_j)$: $Z_i = \sum_b s_b \Delta\tau_b(p_j)$. I set both instruments equal to zero for all pre-policy periods.

To work as a valid instrument, it must satisfy two conditions: first, the instrument should have a strong first stage. Second, the instrument should meet the exclusion restriction. Starting with the first condition, the change in the SDLT should have a strong effect on house prices. More precisely, in my case, I want the change in the tax rate to affect not only house prices mechanically but also change the neighborhood composition of poor versus rich households via a demand channel. The intuition is that neighborhoods with a high share of houses for which the tax rate decreased after the policy should see an increase in demand as the neighborhood becomes more affordable. Thus, we should expect more low-income households to move into this neighborhood. Similarly, areas in which the tax burden on average increased should see an increase in the density of high-income households. The literature in urban economics studying the UK's SDLT confirms that this form of transaction taxes indeed moves house prices significantly. Both Besley et al. (2014) and Best and Kleven (2018) find that a reduction in the transaction tax due to a tax holiday reduced post-sale house prices significantly. On top of that, and importantly for my case, Hilber and Lyytikäinen (2017) show that the SDLT also affects households' mobility and sorting responses. I can confirm the relevance condition in my data by looking at the first stage F-test, which exceeds conventional thresholds.

The second assumption for the instrument to be valid is that the exclusion restriction, which requires that the SDLT reform affects amenity provision only through its effect on house prices. Two considerations support this assumption: (1) The reform was a national policy yielding revenue to national government, not local authorities, making it unlikely to be strategically designed to favor particular neighborhoods. (2) The reform was announced in autumn and implemented in December 2014 with insufficient time for local authorities to anticipate and alter amenity investment plans.

I estimate the following two-stage least squares regressions (one for each endogenous regressor), where the first stages are given by

$$\begin{aligned} \log(\text{ mean price })_i &= \pi_0 + \pi_1 Z_i + X_i'\pi + \gamma_{g(i)} + \varepsilon_i \\ \log(\text{ mean price })_i^2 &= \tilde{\pi}_0 + \tilde{\pi}_1 Z_i^2 + X_i'\tilde{\pi} + \tilde{\gamma}_{g(i)} + \tilde{\varepsilon}_i \end{aligned} \tag{3}$$

where Z_i is one of the instruments explained above, and the vector of controls includes population density, distance to CBD, and distance to town halls, property age, and an urban dummy, and I control for local authority-specific time trends. The second stage is then given by

$$Y_i = \beta_0 + \beta_1 \log(\widehat{\text{mean price}})_i + \log(\widehat{\text{mean price}})_i^2 + X_i'\beta + \gamma_{g(i)} + u_i \tag{4}$$

where Y_i is an accessibility measure to a given public amenity or the accessibility-specific amenity indices, and $\widehat{\text{mean price}}$ is the fitted value from the first stage. Controls and fixed effects are the same as in the first stage, and I cluster standard errors at the local authority level.

Tables 4 and 5 present the results of the 2SLS regressions using the exposure instrument. The resulting coefficients are statistically significant and have the same sign as the ones from the OLS regression. The 2SLS coefficients are 2–3 times larger in absolute magnitude than the OLS coefficients. For example, the OLS coefficient on log house prices for distance to library (Table 2, Column 1, Panel A) is 0.611, while the 2SLS coefficient (Table 4) is 1.715.

There are two potential explanations for why 2SLS coefficients exceed OLS estimates. The first is measurement error in the independent variable, leading to attenuation bias. Secondly, heterogeneous treatment effects. 2SLS identifies the effect only for complier neighborhoods. Those near SDLT thresholds whose prices responded to the reform. The largest amenity gains occur at both distribution tails: in poorer areas with a high share of compliers, increased affordability may have prompted targeted social infrastructure investment; in richer neighbourhoods with a high share of compliers, higher transaction costs concentrated affluent households whose political power drove amenity upgrades. LATE thus averages these strong tail effects, exceeding OLS's more muted population-average relationship.

I present evidence on the relationship between the quality of public amenities and neighbourhood affluence in Appendix Section D.

5 Mechanisms

The U-shaped relationship is surprising: why do both poor and rich neighborhoods have better access, while middle-income neighborhoods are disadvantaged? I examine two broad explanations: Firstly, public amenities might be located in areas where they are used the most (Church and ReVelle, 1976), so the allocation that we observe represents an efficient allocation. Secondly, the literature has shown that political economy plays an important role in the distribution of public funds and public goods (Baskaran and Lopes Da Fonseca, 2021; Grossman and Helpman, 1996; Hodler and Raschky, 2014; Larcinese et al., 2006; Ouasbaa et al., 2025; Solé-Ollé and Sorribas-Navarro, 2008). Therefore, neighborhood affluence could translate into political power, helping these neighborhoods attract more public goods.

5.1 Public amenity usage

The observed U-shape might be explained by the fact that rich and poor people have similar tastes for public amenities. In that case we would observe the same shape in their preferences for public amenities. To run this analysis, I need public amenity usage by type of user. While this data does not exist, I use public library usage as a proxy variable. The Understanding Society Survey is a nationally representative longitudinal survey of UK households. I use three waves from 2011, 2014, and 2018 to. Parents and their children can be matched via unique identifiers, allowing me to link children's library usage to parental income. In the youth questionnaire, respondents are asked how often they visit a public library,[4] with answers being: most days, at least once a week, at least once a month, several times a year, once a year or less, and never/almost never. I code library usage as a binary indicator equal to one if the child visits a public library at least weekly. From the parental survey, I create deciles of parental net weekly household income and estimate the effect of each decile relative to the median (5th) decile. After matching the information on parental income with the children's public library usage, I run the following regression

$$Y_{it} = \sum_{j} \beta_j D_{it,j} + \gamma_t + \epsilon_{it} \tag{5}$$

where Y_{it} is a dummy variable equal to one if the child visits a public library at least once per week, $D_{it,j}$ are indicator variables for a child's i parental income in wave t lies in decile j, with the median decile being the reference category, γ_t is a wave fixed effect, and ϵ_{it} represents an idiosyncratic error term.

Figure 4 displays the coefficients β_j for each income decile. The effect is strongest at the extremes: children from the lowest income decile are 2 percentage points more likely to visit libraries weekly (relative to the median), while children from the highest decile are 1.5 percentage points less likely. The pattern is relatively linear across deciles. This pattern might be explained by different reasons. For instance, low-income parents are more time-constrained than high-income parents and thus need to send their children to after-school programs. Or relatedly, children from low-income backgrounds have fewer resources to spend on school books and thus use libraries more often. Heterogeneous demand can explain part of the U-shape. Children from low-income households use libraries more frequently (Figure 4), consistent with public amenities being complements to lower-income households' consumption. However, children from high-income households use libraries less, yet high-income neighborhoods have substantial public amenity provision. This suggests that demand-based sorting alone cannot explain the full U-shape. Political economy mechanisms likely explain the amenity provision in high-income areas.

5.2 Political economy of public amenity allocation

Motivated by the political economy literature on public spending and public good provision as well as by the literature on political inequality (Cagé, 2024), it is reasonable to expect political econ-

[4]Importantly, this excludes their school libraries.

omy to play a role in explaining the empirical pattern of the relationship between public amenity availability and neighborhood affluence. I examine two proxies for political power—voter participation (measured by age composition) and property holder interest in local outcomes (measured by homeownership rates)—to test whether political economy explains the U-shape. The evidence supports a political economy mechanism: public amenities concentrate in high-political-power neighborhoods at both ends of the income distribution.

The two proxies that I am using are the share of people above the age of 65 and the share of homeowners. Barilari et al. (2025) document that voter participation increases substantially with age. In neighborhoods with higher shares of elderly residents, older voters constitute a larger portion of the electorate, increasing politicians' incentive to respond to their preferences. Therefore, neighborhoods with high shares of residents age 65+ should have greater political influence. Secondly, the 'homevoter hypothesis' posits that homeowners, who bear the capitalized costs and benefits of local public goods, are more politically engaged in local politics (Ahlfeldt and Maennig, 2015; Ahlfeldt et al., 2017). Homeowners have incentives to lobby for amenities and public goods that increase property values, while renters—who do not directly benefit from property value appreciation—have weaker preferences for such spending. Neighborhoods with higher homeownership rates, thus, should exert greater political pressure for amenity provision.

To construct these measures, I rely on census data from 2021, which records homeownership as well as age profiles. For every neighborhood, I construct the share of homeowners and the share of people above the age of 65 and assign them into deciles with respect to these variables. Similarly, I create house price deciles as measures of neighborhood affluence. Figure 5 shows four heatmaps that plot the three-dimensional relationship between political power, neighborhood affluence, and public/private amenity accessibility. On the x-axis of each heatmap are the property transaction price deciles, and on the y-axis, the deciles of the share of people above the age of 65 (Panels (a) and (b)) and the deciles of the share of homeowners (Panels (c) and (d)). I contrast public amenity access (Panels (a) and (c)) to private amenity access (Panels (b) and (d)). Amenity accessibility is measured by the PCA based amenity indices for public and private amenities discussed in section 3.1.

Comparing Panels A and C of Figure 5 (public amenities) to Panels B and D (private amenities), a clear pattern emerges. For public amenities, provision is high in both low-affluence and high-affluence neighborhoods, and this pattern is reinforced in high-political-power areas (high on the y-axis). For private amenities, provision concentrates exclusively in high-affluence, high-power neighborhoods. This suggests that political economy factors specifically drive public amenity provision. In neighborhoods with low political power (high share of residents age 24 and below, Figure 6), public amenities are less available but still show the U-shaped pattern with respect to affluence: some provision in low-affluence neighborhoods (redistribution) and high-affluence neighborhoods (where some wealthy young adults live). Private amenities are concentrated only in high-affluence neighborhoods, showing no responsiveness to low-power neighborhoods regardless of affluence. This absence of private amenities in young neighborhoods reflects that businesses target affluent customers; young-adult areas only attract private amenities if residents are wealthy.

The differential response of public vs. private amenities to neighborhood political power provides evidence that political mechanisms drive public goods provision.[5]

Additionally, I complement this evidence with data from local complaints about neighbourhood issues. The data comes from the online platform FixMyStreet on which citizens can report problems like littering, potholes, graffiti, or noise directly to the relevant authority. The data is available for the years 2007 – 2020 and broken down into seven categories.[6] Figure 7 shows a binned scatterplot of the total number of complaints a neighborhood reported to their relevant authority against its affluence measured as average property prices. The figure suggests that rich neighbourhoods complain significantly more often to their authority than poor neighbourhoods.

Taken together, this presents evidence that to explain the U-shape, one likely needs two mechanisms that work in opposite directions. I show that demand for public amenities can explain why we observe more public amenities in low-income neighborhoods, while political economy mechanisms can explain why we observe more public amenities in high-income neighborhoods.

6 Quantitative Urban Model

Taking stock, the empirical part of this paper has established three facts: first, the relationship between public amenity access and neighbourhood affluence follows a U shape. Second, poor households use public amenities more often than rich households, and third, rich neighbourhoods are more vocal about local issues than poor neighbourhoods. I build a quantitative urban model that incorporate these three facts. The model serves three purposes: (1) quantifying the welfare costs of the observed U-shaped amenity allocation, (2) decomposing the roles of household sorting versus political economy in generating this pattern, and (3) evaluating a counterfactual policy in which political economy plays no role in the allocation of public amenities. The quantification leverages the reduced-form estimates from Section 4 to discipline key parameters.

Households

The city is comprised of $r = 1, ..., S$ residence and $m = 1, .., S$ work locations, also denoted as neighborhoods. Workers ω are heterogeneous in income and can be of either of two types $\theta \in \{l, h\}$, where this corresponds to low and high income. They have preferences over a consumption good $C^{\theta}_{rm}(\omega)$, housing $H^{\theta}_{rm}(\omega)$ and amenities B_r.

The utility function has a Stone-Geary functional form with workers consuming a minimum of housing $\bar{H}$. I follow the literature in choosing a Stone-Geary functional form with workers consuming a subsistence level of housing $\bar{H}$ to prevent workers living on an unrealistically small amount of housing space and to create non-homothetic housing demand (Finlay and Williams,

[5] Because Figure 5 conditions on political power in addition to price, and the underlying age/homeownership data are missing for ∼90% of neighbourhoods, individual cells are estimated on small samples and need not each reproduce the smoother marginal U-shape in Figure 2.

[6] The categories are health, environment, road safety, parking, public space, incivilities, and other.

2025; Gaigné et al., 2022).

$$U_{rm}^{\theta}(\omega) = \frac{z_{rm}(\omega) B_r^{\theta}}{d_{rm}} \left(\frac{C_{rm}^{\theta}(\omega)}{\beta} \right)^{\beta} \left(\frac{H_{rm}^{\theta}(\omega) - \bar{H}}{1-\beta} \right)^{1-\beta}, \qquad H_{rm}^{\theta}(\omega) \geq \bar{H}.$$

where I decompose amenities into a fundamental amenity b_r, like scenic views, a public amenity g_r and a private amenity x_r

$$B_r = b_r g_r^{\eta^{\theta}} x_r^{\rho^{\theta}}$$

The exponents η^{θ} and ρ^{θ} represent the marginal utility of consuming public and private amenities. I allow these marginal utilities to be type-specific, indexed by the superscript θ. This allows for the possibility that it is easier for high-type (high-income) households to substitute between public and private goods.

Plugging this back into the utility function gives

$$U_{rm}^{\theta}(\omega) = \frac{z_{rm}(\omega) b_r \, g_r^{\eta^{\theta}} \, x_r^{\rho^{\theta}}}{d_{rm}} \left(\frac{C_{rm}^{\theta}(\omega)}{\beta} \right)^{\beta} \left(\frac{H_{rm}^{\theta}(\omega) - \bar{H}}{1-\beta} \right)^{1-\beta}$$

I assume the idiosyncratic shocks $z_{rm}(\omega)$ to be distributed according to a Fréchet distribution (type II extreme value) following the literature enabling a closed-form solution for the location choice probabilities (Ahlfeldt et al., 2015; Redding and Rossi-Hansberg, 2017; Redding, 2024). This results in the cumulative distribution function (CDF)

$$F(z_{rm}) = \text{Prob}\left[z_{rm}(\omega) \leq z \right] = e^{-z^{-\varepsilon}}$$

where $\varepsilon > 1$ is the shape parameter that regulates the dispersion of idiosyncratic preferences. This is a simplified distribution compared to Ahlfeldt et al. (2015) as I assume for now no scale parameters that determine the average utility of living in r and working in m respectively. This implies that all systematic differences in the attractiveness of living in r and working in m operate through the observed model objects: residential amenities, wages, commuting costs, and rents. This keeps the location-choice block transparent and avoids introducing additional unobserved shifters, but it also means that the model cannot separately absorb residual residence-workplace-pair attractiveness that is not captured by these fundamentals.

Households earn a wage w_m^{θ} and pay taxes τ on income to pay for the public goods. Households consume housing and the numeraire good. This gives the budget constraint

$$C_{rm}^{\theta}(\omega) + r_r(H_{rm}^{\theta} - \bar{H})(\omega) = w_m^{\theta}(1-\tau)$$

and results in the Lagrangian

$$\mathcal{L} = \frac{z_{rm}(\omega) b_r g_r^{\eta_{\theta}} x_r^{\rho_{\theta}}}{d_{rm}} \left(\frac{C_{rm}^{\theta}(\omega)}{\beta} \right)^{\beta} \left(\frac{H_{rm}^{\theta}(\omega) - \bar{H}}{1-\beta} \right)^{1-\beta} + \lambda \left[(1-\tau) w_m^{\theta} - C_{rm}^{\theta}(\omega) - r_r H_{rm}^{\theta}(\omega) \right].$$

Solving the Lagrangian and plugging the optimal solutions back into the utility function gives indirect utility

$$V_{rm}^{\theta}(\omega) = \frac{z_{rm}(\omega) b_r g_r^{\eta_\theta} x_r^{\rho_\theta}}{d_{rm}} \frac{(1-\tau) w_m^{\theta} - r_r \bar{H}}{r_r^{1-\beta}}, \qquad (1-\tau) w_m^{\theta} > r_r \bar{H}.$$

This expression shows that households like to live in neighborhoods r with higher public amenities, higher private amenities and work in locations m where they earn a high net-of-tax wage. They dislike commutes and high rents.

Relying on properties of the Fréchet distribution, we can solve for the probability that a worker ω decides to live in neighborhood r and works in neighborhood m:[7]

$$\pi_{rm}^{\theta} = \frac{\left(b_r g_r^{\eta_\theta} x_r^{\rho_\theta} \left[w_m^{\theta}(1-\tau) - r_r \bar{H}\right]_+\right)^{\varepsilon} \left(d_{rm} r_r^{1-\beta}\right)^{-\varepsilon}}{\sum_{i=1}^{S} \sum_{j=1}^{S} \left(b_i g_i^{\eta_\theta} x_i^{\rho_\theta} \left[w_j^{\theta}(1-\tau) - r_i \bar{H}\right]_+\right)^{\varepsilon} \left(d_{ij} r_i^{1-\beta}\right)^{-\varepsilon}} \equiv \frac{\Phi_{rm}^{\theta}}{\Phi^{\theta}}. \tag{6}$$

From this expression, it is easy to get the probability that a worker ω lives in location r, π_{Rr}, by summing over all possible work locations j and the probability that a worker ω works in location m, π_{Mm}, by summing over all possible residence locations:

$$\pi_{Rr} = \sum_{m=1}^{S} \pi_{rm} = \frac{\sum_{m=1}^{S} \Phi_{rm}}{\Phi}, \pi_{Mm} = \sum_{r=1}^{S} \pi_{rm} = \frac{\sum_{r=1}^{S} \Phi_{rm}}{\Phi} \tag{7}$$

Similarly to the choice probabilities of living in r and working in m, I can derive the conditional probability that a worker commutes to neighborhood m given that they live in r

$$\pi_{rm|r}^{\theta} = \frac{\left(\frac{\left[w_m^{\theta}(1-\tau) - r_r \bar{H}\right]_+}{d_{rm}}\right)^{\varepsilon}}{\sum_{j=1}^{S} \left(\frac{\left[w_j^{\theta}(1-\tau) - r_r \bar{H}\right]_+}{d_{rj}}\right)^{\varepsilon}}. \tag{8}$$

This shows that conditional on living in a neighborhood r, the probability that a worker commutes to some neighborhood m depends on the wage paid in location m, w_m, discounted by the commuting costs d_{rm} ("bilateral resistance") relative to commuting costs weighted wages in all possible locations j ("multilateral resistance"). According to the same logic, the probability that a worker lives in neighborhood r conditional on working in neighborhood m is given by

$$\pi_{rm|m}^{\theta} = \frac{\left(b_r g_r^{\eta_\theta} x_r^{\rho_\theta} \left[w_m^{\theta}(1-\tau) - r_r \bar{H}\right]_+\right)^{\varepsilon} \left(d_{rm} r_r^{1-\beta}\right)^{-\varepsilon}}{\sum_{i=1}^{S} \left(b_i g_i^{\eta_\theta} x_i^{\rho_\theta} \left[w_m^{\theta}(1-\tau) - r_i \bar{H}\right]_+\right)^{\varepsilon} \left(d_{im} r_i^{1-\beta}\right)^{-\varepsilon}}. \tag{9}$$

This conditional probability depends on the relative attractiveness of a residence location r determined by its fundamental amenities (b_r), public amenities (g_r), private amenities (x_r), commuting costs (d_{rm}), and rents (r_r) relative to the attractiveness of living in any other residence location i.

With the help of the commuting probabilities, I can derive the commuter market clearing

[7] See appendix section **??** for a detailed derivation.

condition where the total workplace population (H_{Mm}) is given by

$$H^{\theta}_{Mm} = \sum_{r=1}^{S} \pi^{\theta}_{rm|r} H^{\theta}_{Rr} = \sum_{r=1}^{S} \frac{\left(\frac{\left[w^{\theta}_{m}(1-\tau) - r_r \bar{H}\right]_+}{d_{rm}}\right)^{\varepsilon}}{\sum_{j=1}^{S}\left(\frac{\left[w^{\theta}_{j}(1-\tau) - r_r \bar{H}\right]_+}{d_{rj}}\right)^{\varepsilon}} H^{\theta}_{Rr}. \tag{10}$$

with H_{Rr} being the residence population.

Local council

The local council provides public amenities subject to a balanced budget. The supply of local public goods is modeled following Maskin and Tirole (2019), who study pork-barrel in local public good provision. The functional form of the utility function of the council follows Grossman and Helpman (1994).

Public amenities are financed through a flat tax rate τ on labor income. Thus, total tax revenue is given by

$$T = \tau \sum_{\theta} \sum_{m=1}^{S} w^{\theta}_{m} H^{\theta}_{Mm}. \tag{11}$$

The local council faces a linear cost to provide the public amenities $c(g) = \xi g_r$ where ξ is the conversion of one unit of the public amenity into monetary units. Thus, the budget constraint is $\sum_{r=1}^{S} \xi g_r = T$.

Define the share of high-type workers in a given neighborhood r as $s_{rH} = \frac{H^{H}_{Rr}}{H_{Rr}}$ and the share of low type in a neighborhood r as $s_{rL} = 1 - s_{rH} = \frac{H^{L}_{Rr}}{H_{Rr}}$ with H^{θ}_{Rr} denoting the number of workers of type θ with residence in neighborhood r. Then the average utility of a given type in neighborhood r is given by

$$\bar{U}^{\theta}_{r}(g_r) = \frac{1}{H^{\theta}_{Rr}} \sum_{m} H^{\theta}_{Rrm} U^{\theta}_{rm}(g_r)$$

I aggregate these type-specific utilities following the Atkinson (1970) style modeling of inequality:

$$W_r(g_r) = \begin{cases} \left(s_{rL}\left(\bar{U}^{L}_{r}(g_r)\right)^{1-\chi} + s_{rH}\left(\bar{U}^{H}_{r}(g_r)\right)^{1-\chi}\right)^{\frac{1}{1-\chi}}, & \chi \neq 1 \\ \left(\bar{U}^{L}_{r}(g_r)\right)^{s_{rL}}\left(\bar{U}^{H}_{r}(g_r)\right)^{s_{rH}}, & \chi = 1 \end{cases}$$

The parameter χ controls the council's aversion to inequality in utilities. When $\chi = 1$, the council values total welfare; as χ increases, the council increasingly weights poor households' welfare. I will assume $\chi = 1$ going forward. This avoids imposing ad-hoc strong redistributive preferences.

Aggregating over all neighborhoods yields aggregate welfare

$$W(g) = \sum_{r} \frac{H_{Rr}}{N} W_r(g_r)$$

where $N = \sum_{r=1}^{S} H_{Rr}$ is the total population.

Motivated by the reduced form evidence, I model the council's utility function following the functional form of Grossman and Helpman (1994). The policymaker maximizes a weighted sum of a political component and a welfare component:

$$U^p(g) = (1-\gamma)P(g) + \gamma W(g), \tag{12}$$

where $P(g)$ denotes the political value of public amenity provision and $W(g)$ the welfare value. $\gamma \in [0,1]$ captures the relative weight on the welfare component. The political component is

$$P(g) = \sum_{r=1}^{S} p(s_{rH}) g_r^{\psi}, \tag{13}$$

where $\psi < 1$ controls the curvature in the political value of public amenities and assures decreasing returns to scale to prevent the allocation from exploding.

The function $p(s_{rH})$ captures political influence as a function of neighborhood composition. From the political economy literature, we know that individuals with larger resources are overrepresented and participate more in politics and thus might be favored in the policy-making process (Brady et al., 1995; Chattopadhyay and Duflo, 2004; Schafer et al., 2022; Elsässer and Schäfer, 2023; Cagé, 2024). Furthermore, to be able to rationalize the empirical U-shape, at least one of the components needs to be non-linear in the public amenity g_r as otherwise the summation of two monotonic functions would itself lead to a monotonically increasing or decreasing function. Therefore I adjust the functional form of Grossman and Helpman (1994), and assume the non-linearity to be in the political component which is thus given as

$$p(s_{rH}) = \alpha_L s_{rL}^2 + \alpha_H s_{rH}^2$$

where α_L and α_H denote the political power of each group of worker type. From the reduced form evidence on the number of complaints in figure 7 by neighbourhood affluence we get $\alpha_H > \alpha_L > 0$.

This allows me to write the council's problem as

$$\max_{\{g_r\}_{r=1}^{S}} \left\{ (1-\gamma) \sum_{r=1}^{S} p(s_{rH}) g_r^{\psi} + \gamma W(g) \right\} \tag{14}$$

subject to

$$\sum_{r=1}^{S} \xi g_r = T. \tag{15}$$

The council maximizes the amount of pubic amenities in each neighbourhood r by weighing off the political component and the welfare component subject to a budget constraint.

Solving the Lagrangian gives the first-order condition for each neighborhood r

$$(1-\gamma)\psi p(s_{rH}) g_r^{\psi-1} + \gamma \frac{\partial W(g)}{\partial g_r} = \lambda \xi. \tag{16}$$

Since aggregate welfare is separable across neighborhoods,

$$\frac{\partial W(g)}{\partial g_r} = \frac{H_{Rr}}{N} \frac{\partial W_r(g_r)}{\partial g_r}. \tag{17}$$

As public amenities enter type-specific utility through $g_r^{\eta_\theta}$, average type-specific utility is proportional to $g_r^{\eta_\theta}$. It follows that

$$\frac{\partial W_r(g_r)}{\partial g_r} = W_r(g_r) \frac{\bar{\eta}_r}{g_r}, \tag{18}$$

where I denote $\bar{\eta}_r = s_{rL}\eta_L + +s_{rH}\eta_H$ as the composition-weighted public-amenity elasticity. The government first-order condition can therefore be written as

$$(1-\gamma)\psi p(s_{rH}) g_r^{\psi-1} + \gamma \frac{H_{Rr}}{N} W_r(g_r) \frac{\bar{\eta}_r}{g_r} = \lambda \xi.$$

This condition shows that the policy maker equalizes the marginal value of public amenities across neighborhoods. The marginal value has two parts. The political term depends on $p(s_{rH})$, while the welfare term depends on population, local welfare, and the composition-weighted public-amenity elasticity $\bar{\eta}_r$. Since we know form the usage of public libraries by types that $\eta_L > \eta_H$, the welfare channel gives relatively more weight to low-type neighborhoods. The political channel gives relatively more weight to neighborhoods with high political influence. The interaction of these channels can generate a U-shaped allocation of public amenities across neighborhood composition.

Production

Workplace employment is determined by the commuting choices of residents. For every worker of type θ, employment in workplace location m is given by

$$H_{Mm}^{\theta} = \sum_{r=1}^{S} \pi_{rm|r}^{\theta} H_{Rr}^{\theta}. \tag{19}$$

and total employment in m thus $H_{Mm} = \sum_{\theta} H_{Mm}^{\theta}$.

In the simplest form of the model, I assume that wages are given exogenously for each type-location pair w_m^{θ}. This allows me to focus on the interaction of public amenities, household sorting, and housing costs.

I include production to determine commercial floor space demand. Competitive firms produce the numeraire good using labor and commercial floor space. Production is Cobb-Douglas with a single labor input implicitly assuming that high- and low-skilled workers are perfect substitutes:

$$Y_m = A_m H_{Mm}^{\alpha} L_{Cm}^{1-\alpha}, \tag{20}$$

where A_m is local fundamental productivity, H_{Mm} is workplace employment, and L_{Cm} denotes

labor demand. As wages are endogenous, I use the production function to account for commercial floor-space demand. Following from the Cobb-Douglas functional form and perfect competition, the local wage bill is given by

$$WB_m = \sum_\theta w_m^\theta H_{Mm}^\theta = \alpha Y_m. \tag{21}$$

Thus, the remaining share of output is spent on commercial floor space $r_m L_{Cm} = (1-\alpha)Y_m$. Combining both gives commercial floor space demand:

$$L_{Cm} = \frac{1-\alpha}{\alpha}\frac{WB_m}{r_m}. \tag{22}$$

Floor space market

The floor space market follows Ahlfeldt et al. (2015) with a fixed supply of floor space in every location, L_r. Floor space is used for either commercial or residential purposes. I denote residential floor space demand by L_{Rr} and commercial floor space demand by L_{Cr}. The floor space market clears whenever

$$L_r = L_{Rr} + L_{Cr} \tag{23}$$

I deviate from Ahlfeldt et al. (2015) in that residential floor space demand follows from the Stone-Geary preferences chosen to incorporate non-homothetic housing demand. Let $Y_m^\theta \equiv (1-\tau)w_m^\theta$ be the after-tax labor income for worker θ working in location m. Then, from the worker's first order condition, optimal housing demand is given by

$$H_{rm}^{\theta *} = \beta \bar{H} + \frac{(1-\beta)Y_m^\theta}{r_r}, \tag{24}$$

where r_r is the price and $\bar{H}$ the subsistence level of floor space. To get the total residential floor space demand in neighbourhood r, we can aggregate this expression over all workplace locations and worker types:

$$L_{Rr} = \sum_\theta \sum_m \pi_{rm|r}^\theta H_{Rr}^\theta \left(\beta \bar{H} + \frac{(1-\beta)Y_m^\theta}{r_r}\right). \tag{25}$$

Defining average after-tax income for worker type θ in location r as $\bar{y}_r^\theta = \sum_m \pi_{rm|r}^\theta Y_m^\theta$, we can rewrite residential floor space demand as

$$L_{Rr} = \beta \sum_\theta H_{Rr}^\theta \bar{H} + \frac{1-\beta}{r_r}\sum_\theta H_{Rr}^\theta \bar{y}_r^\theta. \tag{26}$$

where the first term captures the floor space needed to satisfy the Stone-Geary housing subsistence level. The second term captures the demand for housing above the minimum subsistence level. It increases with workers' disposable income and decreases in floor space price.

Commercial floor space demand is derived from the Cobb-Douglas production function. As

detailed above, wages are taken as exogenous in the baseline quantification. There, given workplace employment H^{θ}_{Mm} and wages w^{θ}_{m}, the local wage bill is given by

$$WB_m = \sum_{\theta} w^{\theta}_{m} H^{\theta}_{Mm}. \tag{27}$$

From the Cobb-Douglas production function with a labor share α, zero profit implies that the total wage bill equals a share of α of local revenue Y_m: $WB_m = \alpha Y_m$. The remainder is then paid to commercial floor space:

$$r_m L_{Cm} = (1-\alpha) Y_m = \frac{1-\alpha}{\alpha} WB_m. \tag{28}$$

This allows us to derive commercial floor space demand as

$$L_{Cm} = \frac{1-\alpha}{\alpha} \frac{WB_m}{r_m}. \tag{29}$$

And total floor space demand in a neighborhood i as

$$L_i = \beta \sum_{\theta} H^{\theta}_{Ri} \bar{H} + \frac{1-\beta}{r_i} \sum_{\theta} H^{\theta}_{Ri} \bar{y}^{\theta}_{r} + \frac{1-\alpha}{\alpha} \frac{WB_i}{r_i}. \tag{30}$$

which we can solve for the equilibrium floor space price

$$r_i = \frac{(1-\beta) \sum_{\theta} H^{\theta}_{Ri} \bar{Y}^{\theta}_{i} + \frac{1-\alpha}{\alpha} WB_i}{L_i - \beta \sum_{\theta} H^{\theta}_{Ri} \bar{H}}. \tag{31}$$

When $\bar{H} = 0$, the expression collapses to the Cobb-Douglas residential housing-demand case.

Equilibrium

Given fundamentals $\{b_r, x_r, L_r\}_{r=1}^{S}$, commuting costs $\{d_{rm}\}_{r,m=1}^{S}$, exogenous type-location wages $\{w^{\theta}_{m}\}_{m=1,\theta\in\{L,H\}}^{S}$, total type populations $\{N^{\theta}\}_{\theta\in\{L,H\}}$, and model parameters, an equilibrium is a set of public amenities, rents, choice probabilities, residence populations, workplace populations, and floor-space allocations

$$\left\{ g_r, r_r, \pi^{\theta}_{rm}, H^{\theta}_{Rr}, H^{\theta}_{Mm}, L_{Rr}, L_{Cr} \right\}_{r,m,\theta}$$

such that the following conditions hold.

First, households choose residence-workplace pairs according to the Fréchet choice probabilities implied by indirect utility,

$$\pi^{\theta}_{rm} = \frac{\Phi^{\theta}_{rm}}{\sum_{i=1}^{S} \sum_{j=1}^{S} \Phi^{\theta}_{ij}}, \tag{32}$$

where Φ^{θ}_{rm} is the deterministic component of indirect utility raised to the Fréchet parameter ε.

Second, residence populations are consistent with household choices:

$$H^{\theta}_{Rr} = N^{\theta} \sum_{m=1}^{S} \pi^{\theta}_{rm}. \tag{33}$$

Third, workplace employment is consistent with commuting choices:

$$H^{\theta}_{Mm} = \sum_{r=1}^{S} \pi^{\theta}_{rm|r} H^{\theta}_{Rr}. \tag{34}$$

Fourth, wages are taken as an exogenous type-location schedule in the focal calibration, while Cobb-Douglas production maps workplace employment and wages into commercial floor-space demand:

$$L_{Cm} = \frac{1-\alpha}{\alpha} \frac{WB_m}{r_m}, \qquad WB_m = \sum_{\theta} w^{\theta}_m H^{\theta}_{Mm}. \tag{35}$$

Fifth, rents clear the local floor-space market in every location:

$$L_r = L_{Rr} + L_{Cr}. \tag{36}$$

Sixth, the local council chooses public amenities optimally. For every location with an interior allocation, public amenities satisfy

$$(1-\gamma)\psi p(s_{rH}) g_r^{\psi-1} + \gamma \frac{H_{Rr}}{N} W_r(g_r) \frac{\bar{\eta}_r}{g_r} = \lambda \xi, \tag{37}$$

where λ is the multiplier on the public-amenity budget constraint.

Finally, the public-amenity budget clears:

$$\sum_{r=1}^{S} \xi g_r = T. \tag{38}$$

The equilibrium therefore links household sorting, floor-space rents, and public-good provision in a fixed point: public amenities affect residential choices, residential choices affect local composition and rents, and local composition feeds back into the government's optimal allocation of public amenities.

7 Quantification

To bring the theoretical model to the data, I discuss the data and key parameters used in the quantification of the model in this section.

7.1 Geography and Data Inputs

I calibrate the model for the city of Birmingham. Birmingham is the second largest city in the UK, with about 1.2 million people living there. All decisions on the distribution of public amenities

are under the control of Birmingham city council. As in the reduced form analysis, I use LSOAs as my main unit of analysis and label them as the neighborhoods of the city.

As inputs, the model requires data on total population, population by income type by residence and workplace location, floor prices/rents, floor-space stock, and the travel time matrix between residence and work locations. I take data on population from the 2021 UK census and define people with a university degree as the high type, following the reduced form analysis. Floor space prices per square meter follow the method of Ahlfeldt et al. (2023) and are then capitalized into annual values following Gaigné et al. (2022). Lastly, I compute the travel time matrix between all neighborhood pairs in Birmingham using information on the public transport network on trains, trams, and buses. For each neighborhood centroid, I take the nearest public transport station and compute the shortest path using Dijkstra's algorithm (Dijkstra, 1959).[8]

Lastly, I calibrate the housing subsistence floor $\bar{H}$ to match empirically plausible rent burdens. I use the Understanding Society data on rent and household income and compare the implied housing-floor cost shares to the observed distribution of rent-to-income rations. The calibrated parameter corresponds to a floor of 25% of income being paid on rent.

7.2 Externally Set Parameters

I set several parameters outside of the model before the internal calibration. I choose these values either from the literature or from existing data sources. Table 7 summarizes the values and sources of each parameter.

I borrow the labor share in production from Ahlfeldt et al. (2015), which is $\alpha = 0.80$. The household consumption share on the consumption good and the Fréchet dispersion parameter come from Heblich et al. (2020) and are set to $\beta = 0.75$ and $\varepsilon = 6.83$, respectively. Lastly, type specific wages are constructed from the production zero-profit condition and normalized to a £35,000 annual wage benchmark. I then impose a constant high-skill premium of two, motivated by recent evidence from England (Britton et al., 2019). Finally, the focal specification sets $\rho_L = \rho_H = 0$, so private amenities x_r are included in the theoretical model but shut down in the quantitative exercise.[9]

7.3 Auxiliary Moments and Targeted Calibration

To quantify the remaining parameters, I use a targeted calibration where some parameters are disciplined by reduced form evidence and others are chosen to match moments of the observed Birmingham city equilibrium. This ensures that each parameter is tied to the empirics.

Two central parameters in the theoretical model are the type-specific marginal utilities of public amenity consumption η_L and η_H. They govern how strongly each type's utility reacts to changes in the availability of public amenities in their neighborhood. I discipline these two

[8] Dijkstra's algorithm iteratively visits the closest not-yet-finalized node in a weighted network, updating the shortest known distance to its neighboring nodes until the least-cost path to the destination is found.

[9] This is a simplifying assumption for the baseline model. Adding private amenities as a substitute for public amenities and heterogeneous substitution elasticities likely makes the sorting mechanism stronger, as rich workers move to low public but high private amenity neighbourhoods.

parameters with the evidence on public library usage by household type from the Understanding Society Survey presented above. This pattern implies that low-income households use public amenities significantly more often than high-income households. I choose the ratio of the two elasticities to match the ratio of public library usage for the richest and the poorest decile in the reduced form analysis. Furthermore, I constrain both parameters to be between 0 and 1, as they are elasticities. In the baseline calibration, I set $\eta_L = 0.8125$ and $\eta_H = 0.375$. The absolute values for both parameters are chosen to match the model's fit to the observed distribution of households across Birmingham neighbourhoods.

The second set of critical parameters are the political weights of each type in the policymaker's objective function. From the political weight function $p(s_{rH}) = \alpha_L(1-s_{rh})+\alpha_H s_{rH}$, we know the optimal share corresponds to $s^* = \frac{\alpha_L}{\alpha_L+\alpha_H}$, so only the ratio of the two matters. I calibrate this ratio to represent the ratio of the complaints data from FixMyStreet. This ratio of complaints by rich versus poor neighbourhoods is equal to 5.75. I then normalize $\alpha_H = 1$ from which it follows that $\alpha_L = 0.174$.[10]

The last key parameter in my adaption of the Ahlfeldt et al. (2015) style model is the policymaker's relative weight of welfare compared to political economy γ. I pick γ to match the U-shape from the observed public amenity distribution across neighborhoods. In the baseline calibration this corresponds to $\gamma = 0.574$.[11]

I treat parameters that are not disciplined by the data as specification choices. I set them to transparent baseline values and examine robustness to alternative values. I set the curvature of the political amenity term to

$$\psi = 0.5, \tag{39}$$

and use the benchmark welfare aggregator with $\chi = 1$, corresponding to the geometric-mean case. The baseline specification also sets the private amenity elasticities

$$\rho_L = \rho_H = 0, \tag{40}$$

Finally, the linear cost parameter for public amenities is a budget normalization. I choose ξ so that the model budget is consistent with observed public-amenity spending:

$$\xi = \frac{T^{obs}}{\sum_{r=1}^{S} g_r^{obs}}. \tag{41}$$

7.4 Amenity Inversion

Given the data and the parameters described above, I recover the residual amenity fundamental that rationalize observed sorting patterns in the base year. Let λ_r^θ denote the observed share of type-θ residents living in location r. Conditional on wages, floor-space prices, commuting times,

[10] The FixMyStreet ratio measures relative complaint activity, not necessarily how governments react to these complaints. Ideally, one would calibrate these weights using data on whether complaints from rich versus poor neighbourhoods are acknowledged, acted upon, or resolved. Complaint rates may nevertheless capture political voice if residents complain more when they expect to be heard.

[11] In a robustness exercise, I calibrate γ to match the nation-wide U-shape, instead of the Birmingham specific U-shape and show it doesn't change results.

and public amenities, the model implies

$$\lambda_r^\theta = \frac{\left(B_r^\theta\right)^\varepsilon r_r^{-(1-\beta)\varepsilon} CMA_r^\theta}{\sum_{i=1}^S \left(B_i^\theta\right)^\varepsilon r_i^{-(1-\beta)\varepsilon} CMA_i^\theta}, \tag{42}$$

where CMA_r^θ denotes commuting-market access from residence location r:

$$CMA_r^\theta = \sum_{l=1}^S \left[(1-\tau)w_l^\theta - r_r \bar{H}\right]_+^\varepsilon d_{rl}^{-\varepsilon}. \tag{43}$$

This term summarizes the workplace opportunities available to type-θ residents of neighborhood r. For each potential workplace, after-tax earnings are reduced by the cost $r_r\bar{H}$ of satisfying the minimum housing requirement in residence r, and then discounted by the commuting cost to that workplace. Thus, conditional on living in r, workplace access is higher when nearby jobs pay more and when the housing floor leaves workers with more disposable income after rent.

Amenities are identified only up to a type-specific scale normalization. I, therefore, invert normalized residence shares to recover

$$B_r^\theta = \left(\lambda_r^\theta\right)^{1/\varepsilon} r_r^{1-\beta} \left(CMA_r^\theta\right)^{-1/\varepsilon}. \tag{44}$$

The recovered type-specific amenity combines residual residential attractiveness with public and private amenities:

$$B_r^\theta = \bar{B}_r g_r^{\eta_\theta} x_r^{\rho_\theta}. \tag{45}$$

In the baseline calibration, $\rho_L = \rho_H = 0$, so private amenities are shut down in the quantitative exercise. I then recover the common residual amenity fundamental as

$$\bar{B}_r^\theta = \frac{B_r^\theta}{g_r^{\eta_\theta}}. \tag{46}$$

Because the model uses a single common residual amenity fundamental, I aggregate the type-specific residuals into $\bar{B}_r$. This object captures location attractiveness that is common across types and not explained by the observed public amenity measure.

The equilibrium exercise holds $\bar{B}_r$ fixed. Public amenities g_r are then reallocated by the council, and the model traces how this reallocation feeds back into household sorting, floor-space prices, and the spatial distribution of welfare.

7.5 Model fit and results

I assess the quality of the model by comparing the equilibrium outcomes against the corresponding objects in the data. The purpose is to assess whether the calibrated model explains the empirical patterns well.

In Figure 8, I plot four graphs with different model-implied outcomes against their empirical counterparts. Panels a and b show the model-predicted distributions of high- and low-type

residents against the data. Both show a very strong and positive correlation. Accordingly, the correlation of the model versus data share of the high-type in panel c is also strongly positively correlated. This reassures that the model is able to explain sorting patterns across Birmingham.

Most importantly, however, the model is able to generate a U-shape of public amenity provision against the distribution of the high skill population (Panel d). The poorest and the richest areas have the highest amenity provision. The mechanism is two-fold. Poor neighbourhoods receive amenities via the welfare channel. Rich neighbourhoods get public amenities via, first, their higher political power, but second, also via sorting mechanisms. Through the Stone-Geary utility function, the house price mechanism hits poor households especially strongly, pricing them out of attractive high public amenity neighbourhoods.

The model, however, creates a narrower support of neighbourhood composition than observed in the data. This may likely be because the baseline specification makes simplifying assumptions that contain a limited set of sorting forces. Wages are imposed as an exogenous schedule instead of arising from a richer labor-demand system. The model, thus, may understate the extent to which different employment hubs attract different types of workers. Furthermore, the Fréchet shock does not contain residence-workplace scale shifters, which means that unobserved location-pair attractiveness cannot create additional sorting of particular groups. Lastly, private amenities are shut down in the baseline specification. This removes a potentially important force through which higher-income workers sort into neighbourhoods with low public amenity but high private amenity availability. Taken together, these restrictions make the model useful to isolate the political economy story in the public amenity distribution across neighbourhoods. But they also limit its ability to reproduce the full dispersion of neighbourhood composition.

Taken together, these diagnostics show that the calibrated model reproduces the main sorting pattern and generates a public-amenity allocation with the same qualitative U-shape as the data. The remaining differences, especially the weaker fit for type-L residents and the compressed support of s_{rH}, point to margins where the model is intentionally parsimonious, most notably the use of fixed type-specific wages and the exclusion of private amenities in the baseline calibration.

Figure 9 plots three maps of Birmingham, mapping the high skill share, the public amenities, and the floor prices. The central object in these maps is not merely the level of g_r, but its spatial alignment with endogenous composition: the model predicts high public amenity provision both in poor neighbourhoods, where welfare gains from amenities are large, and in rich neighborhoods, where political weight is high. Furthermore, high housing costs put additional pressure on low-type workers, pricing them out of attractive neighbourhoods.

8 Counterfactual Analysis

After calibrating and inverting the structural model, I use it to run counterfactual simulations to understand the impact of different policies. In this section, I explore how the spatial allocation of public amenities changes when the council only considers aggregate welfare in its policy function. Specifically, I set $\gamma = 1$ in equation 12. All other parameters are held fixed. I then compare

the counterfactual equilibrium with the baseline equilibrium. More specifically, I show how the allocation of public amenities across space changes and subsequently how welfare and inequality change. I will first define how I measure welfare and inequality.

Welfare Measurement

To quantify the welfare changes implied by the counterfactual policy, I use the indirect utility implied by the Stone-Geary utility specification. For a worker of type θ, living in location r, let B_r^θ denote the type-specific residential amenity, y_r^θ is expected disposable income obtained by aggregating the income of the workers over workplace locations m, r_r the floor space price, and $\bar{H}$ the subsistence housing consumption level. Then, welfare is defined as

$$U_r^\theta = B_r^\theta \frac{y_r^\theta - r_r \bar{H}}{r_r^{1-\beta}},$$

For descriptive comparisons across locations, I normalize welfare separately within each type group:

$$\widetilde{U}_r^\theta = \frac{U_r^\theta}{\sum_r \omega_r^\theta U_r^\theta},$$

where ω_r^θ are resident-population weights within type group θ. I then compute the welfare gap as

$$\Delta_r^U = \log \widetilde{U}_r^H - \log \widetilde{U}_r^L.$$

where positive values indicate that high-type residents have relatively higher welfare in location r than low-type residents.

In addition, I compute a money-metric welfare change between the baseline equilibrium and the counterfactual equilibrium. Let $U_r^{\theta,0}$ denote baseline utility and let $(B_r^{\theta,1}, r_r^1, y_r^{\theta,1})$ denote the welfare-only counterfactual amenity, price, and disposable income. To attain baseline utility under the counterfactual scenario, a worker of type θ requires income

$$y_r^{\theta,\text{req}} = r_r^1 \bar{H} + \frac{U_r^{\theta,0} (r_r^1)^{1-\beta}}{B_r^{\theta,1}}.$$

The compensating variation is then

$$CV_r^\theta = y_r^{\theta,1} - y_r^{\theta,\text{req}}.$$

Thus, $CV_{r\theta} > 0$ means that type θ residents in location r are better off in the welfare-only counterfactual scenario than in the baseline. Conversely, $CV_{r\theta} < 0$ is the compensation required under the counterfactual to restore baseline utility. For scale-free comparisons, I also compute $CV_r^\theta / y_r^{\theta,0}$, the compensating variation relative to baseline disposable income.

Inequality Measurement

To measure inequality in public amenity access and welfare, I compute a Gini coefficient for both. For an outcome x_r and nonnegative weights w_r, locations are ordered from lowest to highest x_r. Let

$$X_k = \frac{\sum_{j \leq k} w_j x_j}{\sum_j w_j x_j}, \qquad W_k = \frac{\sum_{j \leq k} w_j}{\sum_j w_j}$$

denote the cumulative outcome and population shares along the Lorenz curve. The weighted Gini coefficient is

$$G = 1 - \sum_k (X_k + X_{k-1})(W_k - W_{k-1}),$$

with $X_0 = W_0 = 0$. Higher values correspond to higher inequality and $G = 0$ indicates perfect equality.

For public amenity access, the outcome is the model-predicted public amenity g_r and I calculate three different Gini coefficients for total population meaning between type-group inequality, and within type-group inequality for both groups. The same procedure is applied for group welfare defined as above.

8.1 Results

In Figure 10, I present four outcomes of the model-implied counterfactual scenario. Panel A shows the spatial distribution of the public amenity across Birmingham neighbourhoods. If the council only cares about maximizing aggregate welfare, it distributes public amenities more to the northeast of the center and to the southwest border. Comparing this to the distribution of household composition in panel A of Figure 9, it becomes obvious that these are the areas with a high share of low-type workers. This is consistent with the model because when the council maximizes only aggregate welfare, poor neighbourhoods have a high weight as the marginal utility of consuming public amenities is higher for the low than the high type. Panel B shows the implied log welfare changes for each group and for the aggregate. Because amenities are reallocated more towards poor neighbourhoods, low-type workers experience an increase in welfare, while high-type workers experience welfare losses. As the increase for the low type is higher than the decrease for the high type, consequently, aggregate welfare increases.

Panels C and D address the inequality aspect of the public amenity relocation. In panel C, I show the change in public amenity access inequality as measured by the Gini coefficient, as explained above. Inequality in public amenity access is increasing for both types and also in the aggregate. The increase is sizeable by more than 500% in the aggregate public amenity access inequality. This is expected as now rich neighbourhoods don't have the same good amenity access as before the policy change. However, this does not imply that overall inequality is increasing, because households draw utility not only from public amenities. Thus, panel D shows the changes in overall welfare inequality, as well as for both types. Evidently, from the two lowest bars, the change in the council's utility function to value only welfare decreases welfare inequality by 4%.

Taken together, the counterfactual analysis highlights a tradeoff between inequality in public

amenity access and in overall welfare. If the council cares only about welfare in its policy function, inequality in public amenity access increases as these amenities are distributed towards poor neighbourhoods, representing the higher marginal utility for the low type of consuming public amenities. At the same time, this relocation results in a decrease in welfare inequality within the city.

9 Conclusion

Local public amenities are an integral part of the urban environment. They determine where people want to live and how neighbourhoods and their residents develop. In this paper, I examine the spatial allocation of local public amenities within cities and study how neighbourhood composition and political incentives shape this allocation.

Using a newly built dataset on the stock and flow of public amenities across England between 2005 and 2024, I document a robust U-shaped relationship between neighbourhood affluence and public amenity access. The poorest and wealthiest neighbourhoods enjoy the best access to public amenities, while middle-income neighbourhoods are systematically underserved. This pattern holds across different amenity types and access measures. To address endogeneity concerns arising from household sorting and amenity capitalisation into house prices, I exploit quasi-experimental variation from the 2014 Stamp Duty Land Tax reform, which provides plausibly exogenous shocks to neighbourhood affluence. The instrumental variable estimates confirm and strengthen the U-shaped relationship. The evidence points to two forces behind this pattern: low-income households rely more on public amenities, while affluent neighbourhoods have greater political influence over public amenity allocation.

I embed these mechanisms into a quantitative urban model with endogenous public amenity provision. Heterogeneous households sort across neighbourhoods based on wages, commuting access, rents, and amenities, while the local council allocates public amenities by trading off political influence against welfare. I calibrate the model to the city of Birmingham and reproduce the U-shape relationship between public amenity access and neighbourhood affluence. In a counterfactual exercise that removes political influence from the council's objective, I find that amenities are relocated toward more deprived neighbourhoods, city-wide welfare increases, and within-city inequality decreases.

These findings provide novel evidence on the role of public amenities in urban inequality. Unequal access to public amenities is not only the result of household sorting; it is also shaped by policy choices that determine where amenities are provided. This insight is important for local governments facing competing demands across neighbourhoods and deciding where to invest scarce public resources

Several questions remain open. The analysis focuses on access and provision, but future work could further distinguish quality, operating hours, and usage intensity, and could study how specific local political institutions mediate the allocation of amenities. These extensions would help clarify when public amenities reduce spatial inequality and when they reproduce it.

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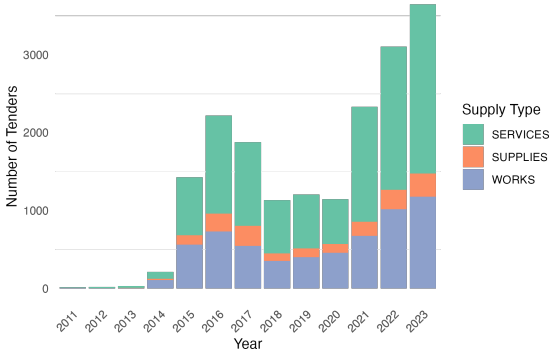
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Figures and tables

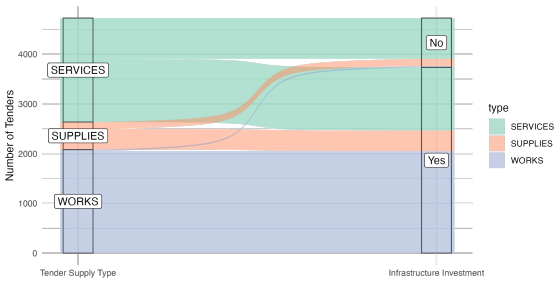
Figures

Figure 1: Procurement tenders and their classification as infrastructure investments

Panel A: Tenders by type over time

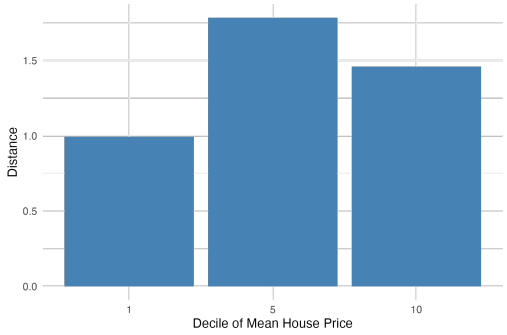


Panel B: Classification into infrastructure investments

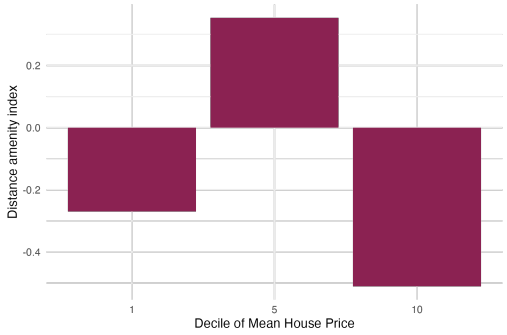


Notes: Panel A shows the number of tenders in every year since the data are available by the type of the tender. These types come from the data. Panel B shows how the tender types translate into infrastructure investments based on the local LLM described in section 3.

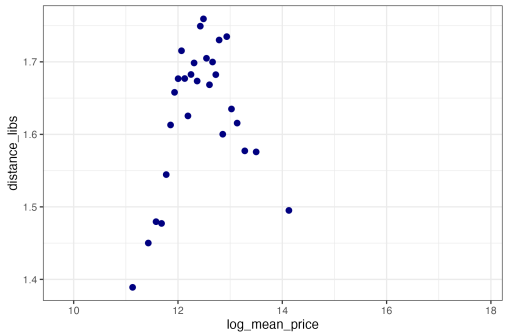
Figure 2: Poorest and richest areas have the best access to public amenities



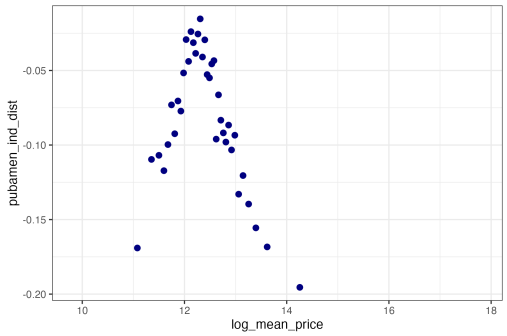
Panel A: Distance to library (mean by decile)



Panel B: Distance-based amenity index (mean by decile)



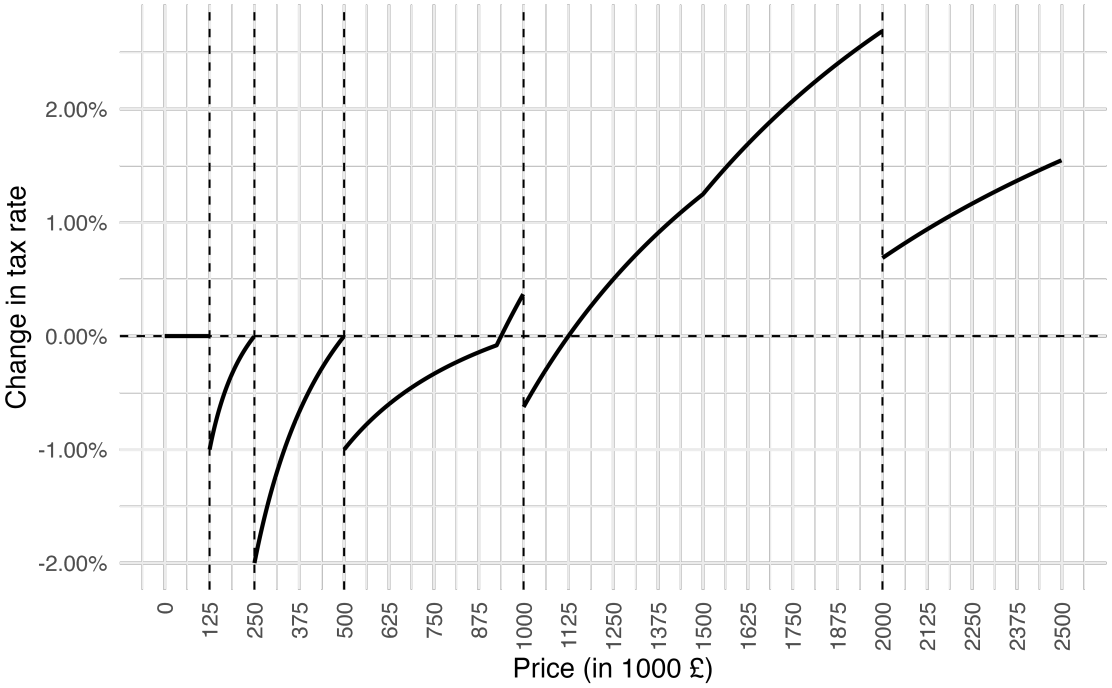
Panel C: Distance to library (binscatter)



Panel D: Distance-based amenity index (binscatter)

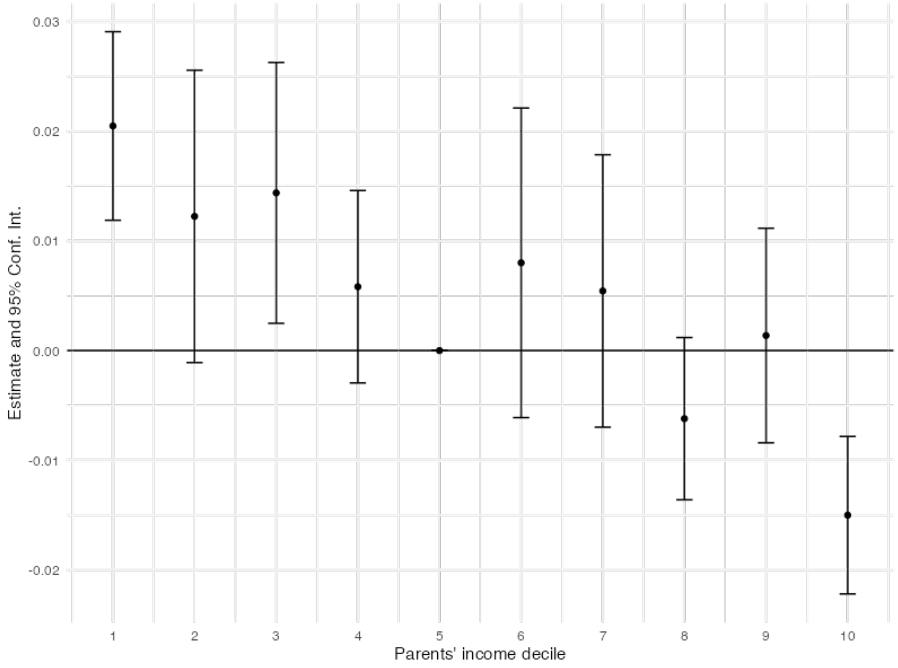
Notes: The figure shows the relationship between amenity access measured as distance to the closest amenity and neighborhood affluence measured as (log) mean property transaction prices. Panels A and B show the average of distance to the closest library and the average distance based amenity index presented in section 3.1 against the first, fifth, and tenth decile of the house price distribution. Panels C and D show binned scatter plots of the same two amenity access variables against a continuous measure of the house price distribution. In each binned scatter plot I control for a LAD fixed effect, distance to CBD, distance to townhouse, population density, age of the property stock, and an urban vs. rural indicator variable.

Figure 3: Stamp duty reform induced change in tax rate across the house price distribution



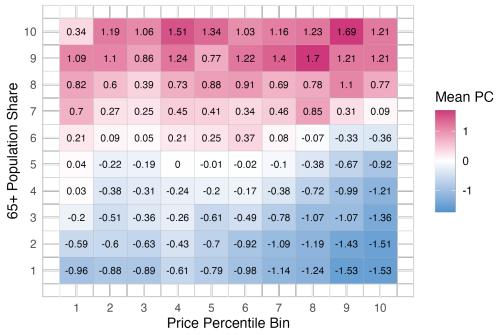
Notes: The figure shows the change in average transaction tax rates for properties of a given value induced by the Stamp Duty Land Tax reform in the UK implemented in December 2014. Source: HMRC; author's calculations, following Ahlfeldt et al. (2024).

Figure 4: Children from low-income background use libraries the most

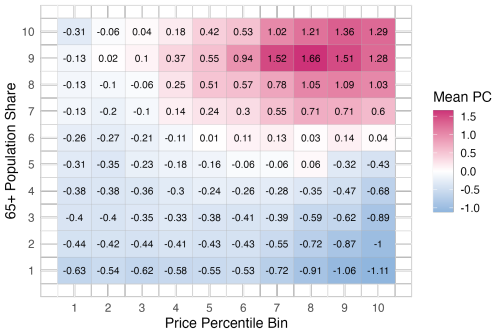


Notes: This figure shows the estimated coefficients and their respective confidence intervals from regression equation 5. The variables are parental household income deciles with the 5th decile denoting the reference category. Data come from the Understanding Society Survey.

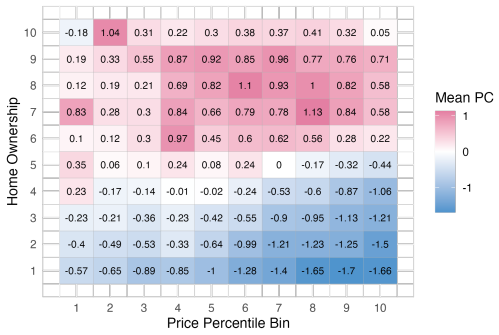
Figure 5: Public amenities in areas of high political power, private amenities in high price areas



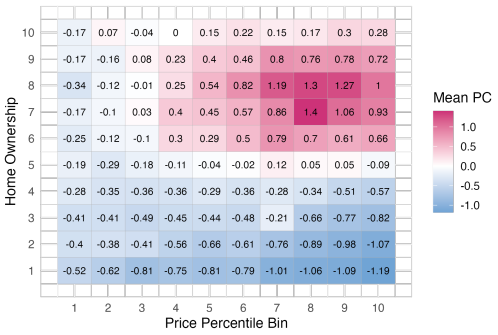
(a) Public amenities and share of old people



(b) Private amenities and share of old people



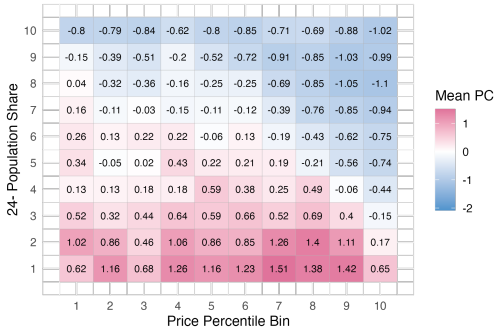
(c) Public amenities and share of homeowners



(d) Private amenities and share of homeowners

Notes: These figures plot heatmaps of average amenity intensity across two dimensions. The x- and y-axis variables are binned into deciles.

Figure 6: Less public amenities in areas of low political power, private amenities in high price areas



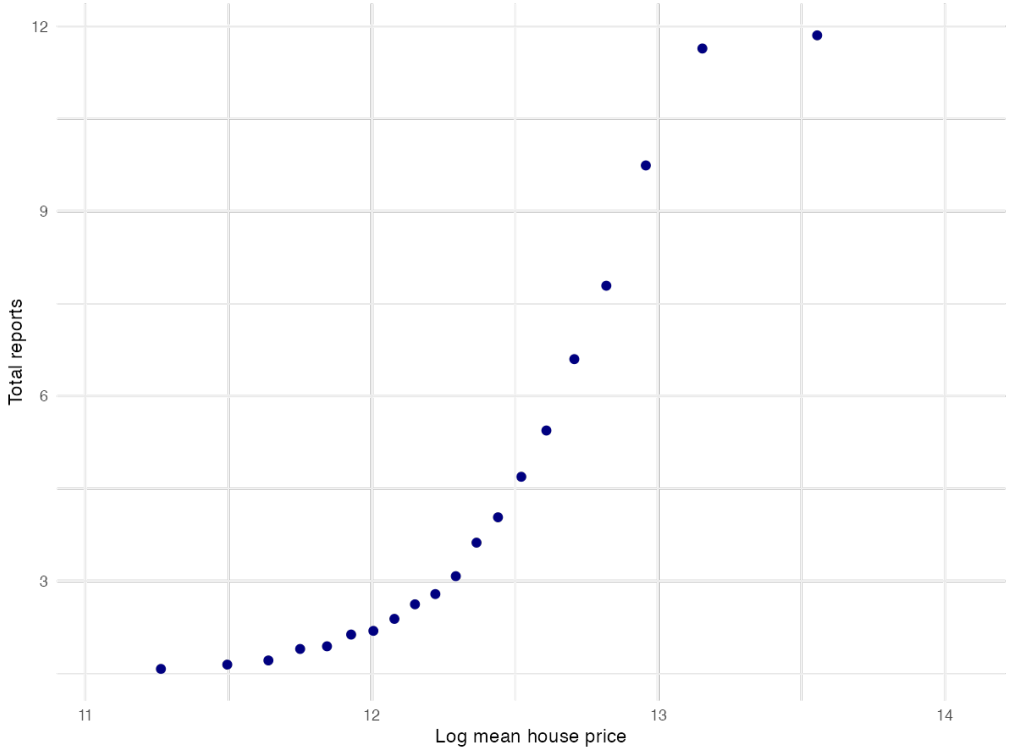
(a) Public amenities and share of young people



(b) Private amenities and share of young people

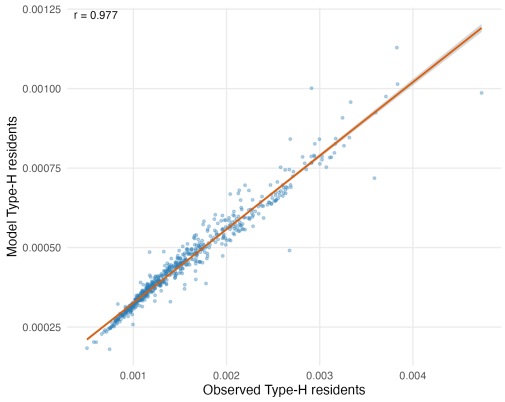
Notes: These figures plot heatmaps of average amenity intensity across two dimensions. The x- and y-axis variables are binned into deciles.

Figure 7: Total complaints increase along the affluence distribution

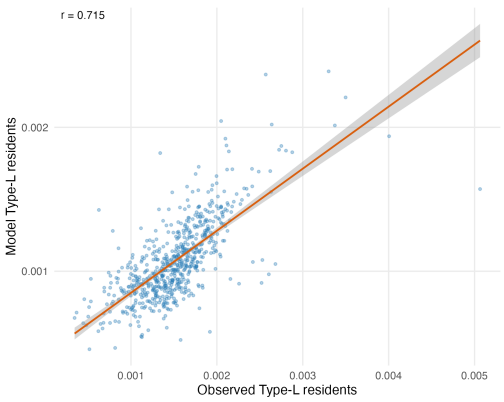


Notes: The figure plots a binscatter of the number of total complaints delivered to a local authority via FixMyStreet against the neighbourhood affluence distribution measured as property prices. The units of observation are neighbourhoods.

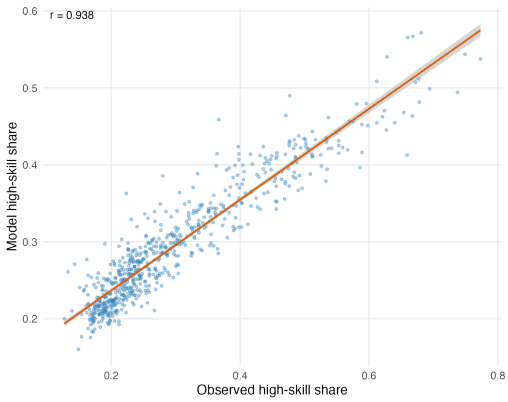
Figure 8: Model Fit



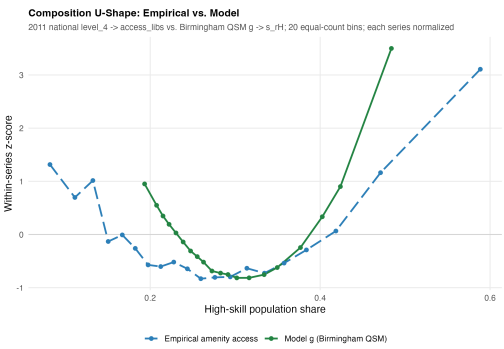
(a) Type-H Residents



(b) Type-L Residents



(c) High-skill share



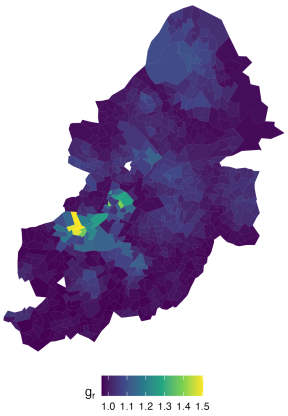
(d) Public amenity U-shape empirical vs. model

Notes: The four panels show the model fit of the quantitative spatial model. In the first three panels, observed normalised population in shares and absolute terms is on the x-axis and the corresponding model prediction on the y-axis. Panel D shows the normalised empirical and model-predicted distribution of public amenities against neighbourhood composition.

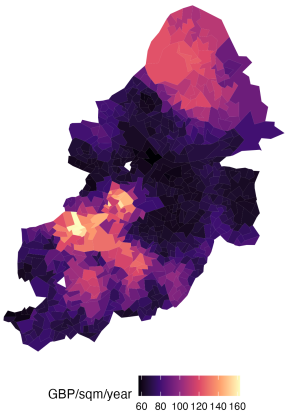
Figure 9: Model Outcomes



(a) High-skill share



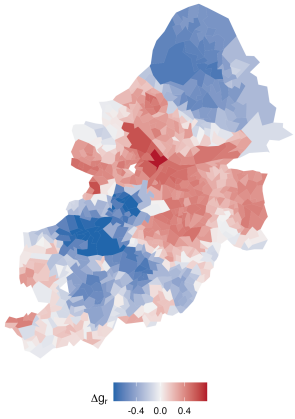
(b) Public amenities



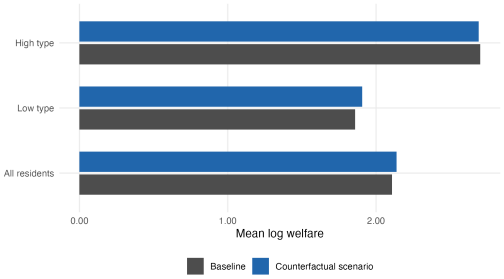
(c) Floor prices

Notes: These maps plot model outcomes of the quantitative spatial model for the city of Birmingham.

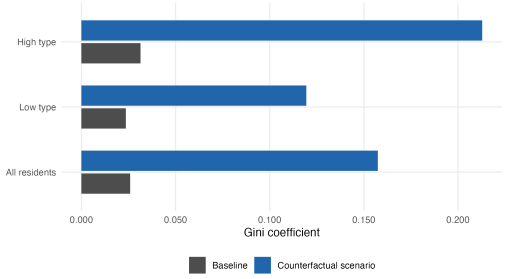
Figure 10: Counterfactual Model Changes



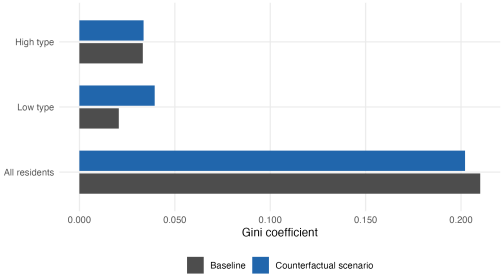
(a) Change in public amenity access



(b) Change in welfare by type



(c) Public amenity inequality (Gini)



(d) Welfare inequality (Gini)

Notes: The four panels present the results from the counterfactual analysis where the council only cares about maximizing welfare, i.e. $\gamma = 1$. Panel A shows the change in public amenity allocation across Birmingham neighbourhoods. Panel B plots the change in welfare, measured by the compensating value, for each type along the neighbourhood composition distribution. Panels C and D plot the baseline and counterfactual inequality in public amenity access and welfare as measured by the compensating value.

Tables

Table 1: Summary Statistics

Variables:	Mean	SD	25th Percentile	75th Percentile	Missing
Distance to library (km)	6.730	13.290	0.870	6.350	0.100
Distance to tender (km)	20.500	32.980	3.560	20.270	0.350
Public amenity index (distance)	0.090	0.830	-0.390	0.280	0.000
Private amenity index (distance)	0.000	0.860	-0.510	0.190	0.450
Libraries within 1km	0.350	0.580	0.000	1.000	0.100
Tenders within 1km	0.280	5.650	0.000	0.000	0.350
Public amenity index (within 1km)	-0.020	0.650	-0.400	0.220	0.000
Private amenity index (within 1km)	0.000	0.970	-0.380	-0.030	0.450
Opening hours nearest library	37.900	14.430	26.500	48.000	0.100
Tender value nearest	1,216,338.580	11,923,715.510	28,635.000	135,045.000	0.590
Mean house price	296,662.530	475,023.360	150,949.680	338,717.210	0.000
Share with university degree	0.304	0.131	0.207	0.380	0.902
Net weekly household income (£)	631.330	141.070	530.000	720.000	0.760
Population density (per km^2)	4,618.460	5,888.680	1,300.810	5,937.110	0.000
Share aged 65+	0.179	0.079	0.120	0.230	0.903
Share homeownership	0.641	0.204	0.496	0.810	0.903
Distance to CBD (km)	3.380	2.440	1.500	4.680	0.000
Distance to townhall (km)	3,834.110	3,240.670	1,527.190	5,146.260	0.000
Urban area	0.829	0.376	1.000	1.000	0.000
Share houses buildt pre 1945	0.630	0.284	0.423	0.876	0.000
IV: exposure	-0.004	0.003	-0.006	-0.002	0.003
IV: average intensity	0.000	0.000	-0.010	0.000	0.000

Table 2: Public libraries and housing prices – **OLS**

Dependent variable	Distance		# 1km		# Presence	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Linear						
Housing value	0.6113***	0.2871**	-0.1458***	-0.0911***	-0.0302***	-0.0440***
	(0.1268)	(0.1307)	(0.0108)	(0.0103)	(0.0033)	(0.0036)
Dependent variable mean	6.7321	6.7321	0.34887	0.34887	0.10073	0.10073
R^2	0.90958	0.91098	0.38482	0.40515	0.01379	0.02024
Observations	605,870	605,870	605,870	605,870	605,870	605,870
Panel B: Adding square						
Housing value	3.6538***	1.8833	-1.0246***	-0.6031***	-0.3579***	-0.3666***
	(1.2668)	(1.2778)	(0.1600)	(0.1488)	(0.0423)	(0.0441)
Housing value square	-0.1208**	-0.0633	0.0349***	0.0203***	0.0130***	0.0128***
	(0.0467)	(0.0470)	(0.0063)	(0.0059)	(0.0016)	(0.0017)
Dependent variable mean	6.7321	6.7321	0.34887	0.34887	0.10073	0.10073
R^2	0.90961	0.91099	0.38615	0.40560	0.01442	0.02084
Observations	605,870	605,870	605,870	605,870	605,870	605,870
Regression specification:						
LAD x Year FE	X	X	X	X	X	X
Controls		X		X		X

Notes: The table presents estimated coefficients for the relationship between public library access and neighborhood affluence, measured as (log) mean transaction prices. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table 3: Amenity index and housing prices – OLS

Dependent variable	Distance		# 1km		# Presence	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Linear						
Housing value	0.1377***	0.0433***	-0.1360***	-0.0438***	0.0047	-0.0198**
	(0.0111)	(0.0065)	(0.0147)	(0.0148)	(0.0091)	(0.0089)
Dependent variable mean	0.08817	0.08817	-0.02282	-0.02282	0.00031	0.00031
R^2	0.85853	0.88230	0.37956	0.42389	0.01684	0.02658
Observations	673,342	673,342	673,342	673,342	673,342	673,342
Panel B: Adding square						
Housing value	0.7949***	0.1863**	-2.0761***	-1.3698***	-1.2553***	-1.1936***
	(0.1060)	(0.0720)	(0.2305)	(0.2421)	(0.2495)	(0.2415)
Housing value square	-0.0260***	-0.0057**	0.0768***	0.0524***	0.0499***	0.0464***
	(0.0040)	(0.0027)	(0.0092)	(0.0097)	(0.0100)	(0.0097)
Dependent variable mean	0.08817	0.08817	-0.02282	-0.02282	0.00031	0.00031
R^2	0.85889	0.88232	0.38473	0.42626	0.01918	0.02857
Observations	673,342	673,342	673,342	673,342	673,342	673,342
Regression specification:						
LAD x Year FE	X	X	X	X	X	X
Controls		X		X		X

Notes: The table presents estimated coefficients for the relationship between public amenity access, measured as the amenity index outlined in 3.1, and neighborhood affluence, measured as (log) mean transaction prices. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table 4: Public libraries and housing prices – **2SLS** transaction tax instrument

Dependent variable	Distance		# 1km		# Presence	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Linear						
Housing value	1.7145***	0.9742	-0.4248***	-0.3179***	-0.2005***	-0.2734***
	(0.5583)	(0.7084)	(0.0680)	(0.0690)	(0.0407)	(0.0625)
Dependent variable mean	6.7321	6.7321	0.34887	0.34887	0.10073	0.10073
R^2	0.90854	0.91062	0.34994	0.38479	-0.03042	-0.05070
Observations	605,870	605,870	605,870	605,870	605,870	605,870
F-test (1st stage), Housing value	31,763.5	20,851.1	31,763.5	20,851.1	31,763.5	20,851.1
Panel B: Adding square						
Housing value	5.0571***	3.3729	-1.0386***	-0.9265***	-0.5621***	-0.8559***
	(1.6780)	(2.2216)	(0.2333)	(0.2324)	(0.1133)	(0.1381)
I(log_mean_price$\hat{2}$)	-0.1618***	-0.1086	0.0297***	0.0261***	0.0175***	0.0270***
	(0.0581)	(0.0749)	(0.0092)	(0.0089)	(0.0045)	(0.0051)
Dependent variable mean	6.7321	6.1113	0.34887	0.36429	0.10073	0.10087
R^2	0.90949	0.90941	0.37675	0.37629	0.00169	-0.00535
Observations	605,870	467,795	605,870	467,795	605,870	467,795
F-test (1st stage), Housing value	35,383.4	13,840.9	35,383.4	13,840.9	35,383.4	13,840.9
F-test (1st stage), I(log_mean_price$\hat{2}$)	31,653.6	13,470.7	31,653.6	13,470.7	31,653.6	13,470.7
Regression specification:						
LAD x Year FE	X	X	X	X	X	X
Controls		X		X		X

Notes: The table presents estimated coefficients for the 2SLS regression of public library access and neighborhood affluence, measured as (log) mean transaction prices. Transaction prices are instrumented using the exposure to the SDLT reform instrument. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table 5: Amenity index and housing prices – **2SLS** transaction tax instrument

Dependent variable	Distance		# 1km		# Presence	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Linear						
Housing value	0.4381***	0.1365***	-0.8220***	-0.6554***	-0.6059***	-0.8030***
	(0.0972)	(0.0505)	(0.1929)	(0.2065)	(0.1760)	(0.2627)
Dependent variable mean	0.08817	0.08817	-0.02282	-0.02282	0.00031	0.00031
R^2	0.83847	0.88060	0.21067	0.30543	-0.12649	-0.18150
Observations	673,342	673,342	673,342	673,342	673,342	673,342
F-test (1st stage), Housing value	22,625.5	13,983.0	22,625.5	13,983.0	22,625.5	13,983.0
Panel B: Adding square						
Housing value	1.2657***	0.3344***	-2.4833***	-1.6883***	-1.9089***	-1.9919***
	(0.1410)	(0.1052)	(0.3245)	(0.3042)	(0.3864)	(0.3539)
I(log_mean_price$\hat{2}$)	-0.0423***	-0.0112***	0.0848***	0.0582***	0.0665***	0.0670***
	(0.0052)	(0.0037)	(0.0134)	(0.0130)	(0.0164)	(0.0149)
Dependent variable mean	0.08817	0.08817	-0.02282	-0.02282	0.00031	0.00031
R^2	0.85795	0.88229	0.36975	0.41696	-0.00185	0.00234
Observations	673,342	673,342	673,342	673,342	673,342	673,342
Regression specification:						
LAD x Year FE	X	X	X	X	X	X
Controls		X		X		X

Notes: The table presents estimated coefficients for the 2SLS regression of public amenity access, measured as the amenity index outlined in 3.1, and neighborhood affluence, measured as (log) mean transaction prices. Transaction prices are instrumented using the exposure to the SDLT reform instrument. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table 6: Infrastructure investments and housing prices – **OLS**, **TWFE**

Dependent variable	Distance		# 1km		# Presence		Investment value	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Linear								
Housing value	3.5632***	3.5384***	0.0710**	0.0739**	0.0369**	0.0369**	-0.0641***	-0.0642***
	(0.8704)	(0.8687)	(0.0347)	(0.0348)	(0.0178)	(0.0178)	(0.0243)	(0.0241)
Dependent variable mean	20.482	20.482	0.27713	0.27713	0.04258	0.04258	11.328	11.328
R^2	0.59236	0.59246	0.31625	0.31630	0.36193	0.36193	0.26230	0.26230
Observations	438,234	438,234	438,234	438,234	438,234	438,234	278,260	278,260
Panel B: Adding square								
Housing value	-47.6188***	-47.8057***	1.3136	1.3350	0.1164	0.1165	-0.0268	-0.0271
	(12.8957)	(12.8869)	(1.3775)	(1.3759)	(0.3278)	(0.3278)	(0.6105)	(0.6109)
Housing value square	1.9573***	1.9634***	-0.0475	-0.0482	-0.0030	-0.0030	-0.0014	-0.0014
	(0.4843)	(0.4839)	(0.0517)	(0.0516)	(0.0121)	(0.0121)	(0.0235)	(0.0235)
Dependent variable mean	20.482	20.482	0.27713	0.27713	0.04258	0.04258	11.328	11.328
R^2	0.59318	0.59328	0.31627	0.31632	0.36193	0.36193	0.26230	0.26230
Observations	438,234	438,234	438,234	438,234	438,234	438,234	278,260	278,260
Turning point	12.165	12.174	13.822	13.841	19.146	19.143	-9.3296	-9.4847
Turning point decile	5	5	10	10	10	10	1	1
Regression specification:								
LSOA FE	X	X	X	X	X	X	X	X
Year FE	X	X	X	X	X	X	X	X
Controls		X		X		X		X

Notes: The table presents estimated coefficients of a regression of infrastructure investments on neighborhood affluence, measured as (log) mean transaction prices. Transaction prices are instrumented using the exposure to the SDLT reform instrument. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

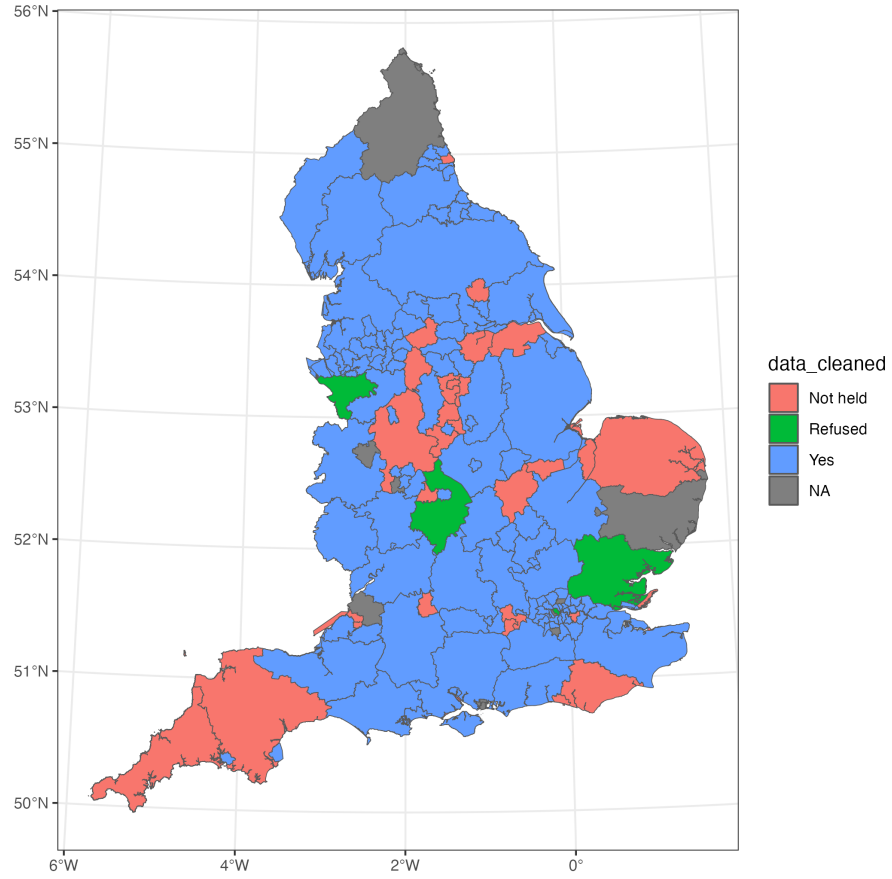
Table 7: Model Parameters and Quantification

Parameter	Description	Value	Source / Moment
A. Directly measured or estimated outside the model			
d_{rl}	Bilateral commuting costs	Data	Travel-time matrix (own computation)
L_r	Fixed floor-space supply	Data	Implied fixed floor-space stock
N^L, N^H	Type populations	Data	UK census (2021), ONS
g_r^{obs}	Baseline public amenities	Data	Public-amenity access index
τ	Public-amenity tax rate	0.20	Average income tax UK (ONS)
α_L	Low-type political weight	0.174	Political-weight calibration using FixMyStreet Complaints
α_H	High-type political weight	1.000	Normalization
η_L/η_H	Relative public-amenity sensitivity	2.17	Calibrated to USOC library-usage gradient
η_L, η_H	Public-amenity elasticities	0.8125, 0.375	Ratio from USOC; common scale chosen to improve type-fit moments
γ	Welfare weight in council objective	0.574	Match empirical amenity U-shape
$\bar{H}$	Stone-Geary housing floor	0.25	Housing expenditure share Birmingham (USOC)
B. Internally calibrated or normalized			
ξ	Unit cost of public amenities	$\frac{T^{obs}}{\sum_r g_r^{obs}}$	Baseline budget normalization
ρ_L, ρ_H	Private-amenity elasticities	0	Focal normalization
C. Taken from literature / standard quantitative urban values			
ε	Fréchet dispersion parameter	5.25	Heblich et al. (2020)
β	Consumption share in household utility	0.75	Heblich et al. (2020)
α	Labor share in production	0.80	Ahlfeldt et al. (2015)
χ	Atkinson welfare-aggregation parameter	1	Geometric-mean welfare benchmark
ψ	Political curvature in g_r	0.5	Diminishing political returns benchmark

Notes: This table reports all parameters used in the quantitative urban model.

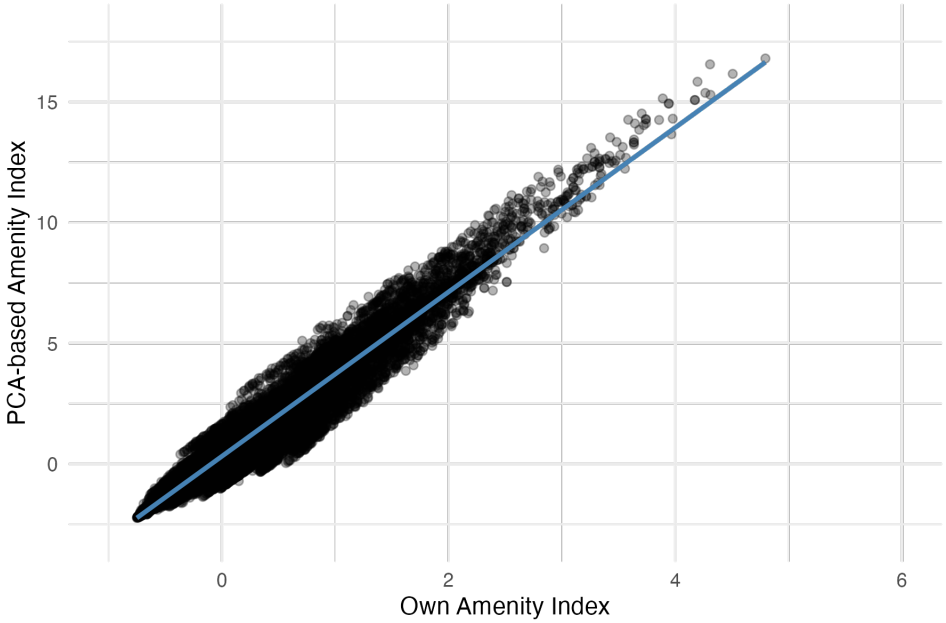
A Appendix Figures

Figure A1: Successful FOI requests



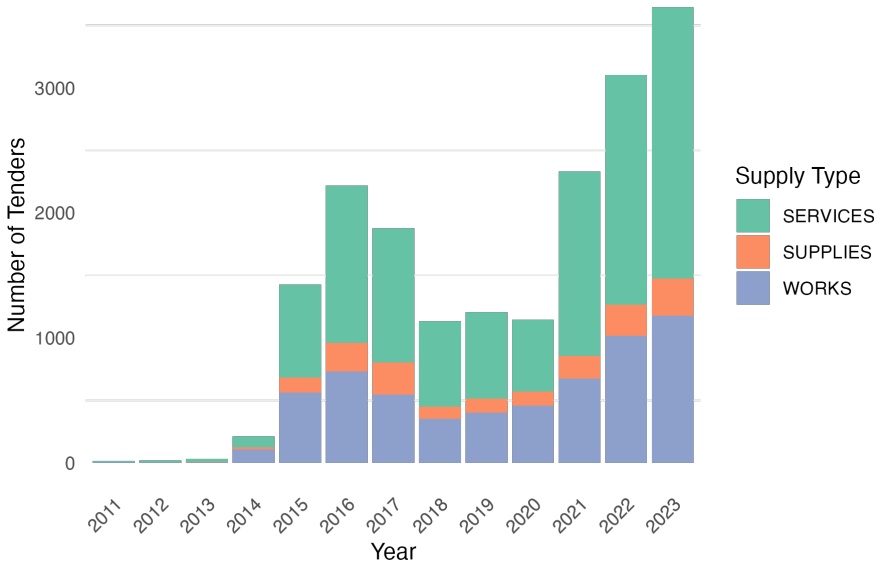
Notes: This figure shows the spatial coverage and success rate of the FOI requests.

Figure A2: Correlation between average and PCA-based amenity indices



Notes: This figure shows the correlation between the two amenity indices that I create.

Figure A3: Procurement tenders by type over time



Notes: This figure shows the temporal variation in the number of tenders by supply type.

B Appendix Tables

B.1 OLS analysis

Table A1: Public libraries and share of people with university degree – **OLS**

Dependent variable	Distance		# 1km		# Presence	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Linear						
Share with degree	1.5428***	0.5295	-0.2954***	-0.1897***	-0.1062***	-0.1546***
	(0.5435)	(0.5678)	(0.0537)	(0.0545)	(0.0161)	(0.0175)
Dependent variable mean	6.5696	6.5696	0.34046	0.34046	0.10090	0.10090
R^2	0.90680	0.90857	0.36408	0.39743	0.01376	0.02014
Observations	66,026	66,026	66,026	66,026	66,026	66,026
Panel B: Adding square						
Share with degree	6.0026***	2.6882	-0.9538***	-0.4081**	-0.2758***	-0.3783***
	(1.7984)	(1.9055)	(0.2057)	(0.1926)	(0.0521)	(0.0505)
Share with degree square	-6.3629***	-3.0578	0.9393***	0.3094	0.2419***	0.3168***
	(1.9143)	(2.0170)	(0.3153)	(0.2973)	(0.0726)	(0.0693)
Dependent variable mean	6.5696	6.5696	0.34046	0.34046	0.10090	0.10090
R^2	0.90689	0.90859	0.36520	0.39755	0.01400	0.02054
Observations	66,026	66,026	66,026	66,026	66,026	66,026
Regression specification:						
LAD x Year FE	X	X	X	X	X	X
Controls		X		X		X

Notes: The table presents estimated coefficients for the relationship between public library access and neighborhood affluence, measured as the share of people with a university degree. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table A2: Public libraries and household income – **OLS**

Dependent variable	Distance		# 1km		# Presence	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Linear						
log_net_weekly_income	-0.1134***	-0.1280***	1.8267***	0.5911***	-0.6266***	-0.3951***
	(0.0116)	(0.0117)	(0.2091)	(0.1401)	(0.0474)	(0.0493)
Dependent variable mean	0.10163	0.10163	2.4828	2.4828	0.46299	0.46299
R^2	0.01499	0.01977	0.78759	0.81800	0.29474	0.33072
Observations	133,331	133,331	133,331	133,331	133,331	133,331
Panel B: Adding square						
log_net_weekly_income	-0.7977**	-0.6639**	8.8831**	1.1905	-5.7758***	-3.2484**
	(0.3138)	(0.3059)	(4.2195)	(3.5739)	(1.4661)	(1.2752)
log_net_weekly_income square	0.0534**	0.0418*	-0.5505*	-0.0467	0.4017***	0.2225**
	(0.0243)	(0.0237)	(0.3279)	(0.2802)	(0.1149)	(0.1003)
Dependent variable mean	0.10163	0.10163	2.4828	2.4828	0.46299	0.46299
R^2	0.01507	0.01982	0.78765	0.81801	0.29590	0.33108
Observations	133,331	133,331	133,331	133,331	133,331	133,331
Regression specification:						
LAD x Year FE	X	X	X	X	X	X
Controls		X		X		X

Notes: The table presents estimated coefficients for the relationship between public library access and neighborhood affluence, measured as (log) average estimated net weekly household income. The unit of observation is an aggregation of neighborhoods (MSOA). Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table A3: Amenity index and share of people with university degree – **OLS**

Dependent variable	Distance		# 1km		# Presence	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Linear						
Share with degree	0.1658***	0.0100	-0.1977***	-0.0721	-0.0283	-0.1535***
	(0.0329)	(0.0222)	(0.0603)	(0.0506)	(0.0209)	(0.0231)
Dependent variable mean	0.26279	0.26279	0.00154	0.00154	0.00794	0.00794
R^2	0.90973	0.92637	0.35744	0.42773	0.02114	0.03640
Observations	66,026	66,026	66,026	66,026	66,026	66,026
Panel B: Adding square						
Share with degree	0.7786***	0.0840	-1.0037***	-0.2654	-0.1722**	-0.4028***
	(0.0850)	(0.0688)	(0.2073)	(0.1759)	(0.0734)	(0.0741)
Share with degree square	-0.8742***	-0.1048	1.1499***	0.2737	0.2053**	0.3531***
	(0.1018)	(0.0822)	(0.3332)	(0.2820)	(0.1032)	(0.1062)
Dependent variable mean	0.26279	0.26279	0.00154	0.00154	0.00794	0.00794
R^2	0.91016	0.92638	0.35953	0.42785	0.02123	0.03664
Observations	66,026	66,026	66,026	66,026	66,026	66,026
Regression specification:						
LAD x Year FE	X	X	X	X	X	X
Controls		X		X		X

Notes: The table presents estimated coefficients for the relationship between public amenity access and neighborhood affluence, measured as the share of people with a university degree. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table A4: Amenity index and household income – **OLS**

Dependent variable	Distance		# 1km		# Presence	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Linear						
Log net weekly income	0.4684***	0.1284***	-0.5847***	-0.3038***	-0.2043***	-0.2242***
	(0.0352)	(0.0243)	(0.0441)	(0.0414)	(0.0199)	(0.0217)
Dependent variable mean	0.07774	0.07774	-0.00710	-0.00710	-0.00619	-0.00619
R^2	0.83839	0.88865	0.36996	0.44581	0.02987	0.04843
Observations	164,595	164,595	164,595	164,595	164,595	164,595
Panel B: Adding square						
Log net weekly income	3.0306***	-0.0070	-5.8635***	-2.6004**	-3.7379***	-2.9362***
	(0.6922)	(0.4504)	(1.1997)	(1.1376)	(0.5457)	(0.5595)
Log net weekly income square	-0.1999***	0.0106	0.4119***	0.1791**	0.2757***	0.2115***
	(0.0538)	(0.0347)	(0.0941)	(0.0892)	(0.0420)	(0.0433)
Dependent variable mean	0.07774	0.07774	-0.00710	-0.00710	-0.00619	-0.00619
R^2	0.83857	0.88865	0.37168	0.44613	0.03083	0.04899
Observations	164,595	164,595	164,595	164,595	164,595	164,595
Regression specification:						
LAD x Year FE	X	X	X	X	X	X
Controls		X		X		X

Notes: The table presents estimated coefficients for the relationship between public amenity access and neighborhood affluence, measured as (log) average estimated net weekly household income. The unit of observation is an aggregation of neighborhoods (MSOA). Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table A5: Public library opening hours and housing prices – **OLS**

Dependent variable	Hours nearest library		Hours within 1km	
	(1)	(2)	(3)	(4)
Panel A: Linear				
Housing value	-1.0620***	0.4686	0.2011	0.5232
	(0.3095)	(0.2902)	(0.4072)	(0.3396)
Dependent variable mean	37.892	37.892	40.036	40.036
R^2	0.39879	0.42393	0.35213	0.42231
Observations	605,870	605,870	182,423	182,423
Panel B: Adding square				
Housing value	-5.8408*	6.5529**	-8.4069**	2.3839
	(3.3216)	(2.9274)	(3.8589)	(3.1139)
Housing value square	0.1897	-0.2412**	0.3392**	-0.0733
	(0.1260)	(0.1122)	(0.1444)	(0.1190)
Dependent variable mean	37.892	37.892	40.036	40.036
R^2	0.39885	0.42403	0.35241	0.42233
Observations	605,870	605,870	182,423	182,423
Regression specification:				
LAD x Year FE	X	X	X	X
Controls		X		X

Notes: The table presents estimated coefficients of a regression of public library quality on neighborhood affluence, measured as (log) mean transaction prices. Transaction prices are instrumented using the exposure to the SDLT reform instrument. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table A6: Public library opening hours and housing prices – **2SLS** Transaction tax instrument

Dependent variable	Hours nearest library		Hours within 1km	
	(1)	(2)	(3)	(4)
Panel A: Linear				
Housing value	-3.8065***	1.6941	-36.5411	-12.8891
	(1.2963)	(1.3804)	(57.6651)	(19.9661)
Dependent variable mean	37.892	37.892	40.036	40.036
R^2	0.39331	0.42296	-0.62710	0.29657
Observations	605,870	605,870	182,423	182,423
Panel B: Adding square				
Housing value	-7.1294	9.7007**	-36.0478***	-13.0895
	(5.0735)	(4.6248)	(13.6939)	(9.7089)
I(log_mean_price$\hat{2}$)	0.1608	-0.4131**	1.1276**	0.3372
	(0.1900)	(0.1764)	(0.4961)	(0.3551)
Dependent variable mean	37.892	37.892	40.036	40.036
R^2	0.39589	0.42307	0.30865	0.40404
Observations	605,870	605,870	182,423	182,423
Regression specification:				
LAD x Year FE	X	X	X	X
Controls		X		X

Notes: The table presents estimated coefficients of a 2SLS regression of public library quality on neighborhood affluence, measured as (log) mean transaction prices. Transaction prices are instrumented using the exposure to the SDLT reform instrument. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table A7: Infrastructure investments and housing prices – **OLS**

Dependent variable	Distance		# 1km		# Presence		Investment value	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Linear								
Housing value	0.4720***	0.1002	0.0069	0.1033*	0.0534***	0.0549***	-0.0137	-0.0178
	(0.0964)	(0.0898)	(0.0465)	(0.0534)	(0.0177)	(0.0168)	(0.0125)	(0.0131)
Dependent variable mean	20.482	20.482	0.27713	0.27713	0.04258	0.04258	11.328	11.328
R^2	0.97882	0.97917	0.08396	0.08662	0.00997	0.01058	0.74086	0.74096
Observations	438,234	438,234	438,234	438,234	438,234	438,234	278,260	278,260
Panel B: Adding square								
Housing value	5.7535***	3.2083**	-2.5838***	-1.6068*	-0.3700	-0.2595	0.2195	0.1933
	(1.3313)	(1.2873)	(0.8524)	(0.8603)	(0.2621)	(0.2747)	(0.1769)	(0.1720)
Housing value square	-0.2073***	-0.1218**	0.1017***	0.0670**	0.0166	0.0123	-0.0092	-0.0083
	(0.0493)	(0.0479)	(0.0327)	(0.0332)	(0.0104)	(0.0108)	(0.0070)	(0.0068)
Dependent variable mean	20.482	20.482	0.27713	0.27713	0.04258	0.04258	11.328	11.328
R^2	0.97884	0.97917	0.08411	0.08669	0.01000	0.01060	0.74087	0.74097
Observations	438,234	438,234	438,234	438,234	438,234	438,234	278,260	278,260
Regression specification:								
LAD x Year FE	X	X	X	X	X	X	X	X
Controls		X		X		X		X

Notes: The table presents estimated coefficients of a regression of infrastructure investments on neighborhood affluence, measured as (log) mean transaction prices. Transaction prices are instrumented using the exposure to the SDLT reform instrument. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table A8: Infrastructure investments and housing prices – **2SLS** Transaction tax instrument

Dependent variable	Distance		# 1km		# Presence		Investment value	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Linear								
Housing value	6.1444	6.2471	-6.2395	-13.1071	-1.2841	-3.0363	0.0808	0.0899
	(5.2743)	(13.0212)	(5.8765)	(28.3137)	(1.2022)	(6.4668)	(0.1089)	(0.1488)
Dependent variable mean	20.482	20.482	0.27713	0.27713	0.04258	0.04258	11.328	11.328
R^2	0.97374	0.97382	-0.12638	-0.75580	-0.05696	-0.30954	0.74036	0.74038
Observations	438,234	438,234	438,234	438,234	438,234	438,234	278,260	278,260
Panel B: Adding square								
Housing value	7.8081***	3.2380	-7.8740***	-7.1187***	-1.6354***	-1.6325***	0.2482	0.2083
	(1.8560)	(1.9782)	(2.2795)	(2.3426)	(0.4957)	(0.5164)	(0.3308)	(0.3170)
I(log_mean_price$\hat{2}$)	-0.2789***	-0.1236*	0.2740***	0.2460***	0.0589***	0.0577***	-0.0096	-0.0083
	(0.0659)	(0.0700)	(0.0793)	(0.0811)	(0.0186)	(0.0191)	(0.0136)	(0.0131)
Dependent variable mean	20.482	20.482	0.27713	0.27713	0.04258	0.04258	11.328	11.328
R^2	0.97883	0.97917	0.07932	0.08194	0.00849	0.00884	0.74086	0.74096
Observations	438,234	438,234	438,234	438,234	438,234	438,234	278,260	278,260
Regression specification:								
LAD x Year FE	X	X	X	X	X	X	X	X
Controls		X		X		X		X

Notes: The table presents estimated coefficients of a 2SLS regression of infrastructure investments on neighborhood affluence, measured as (log) mean transaction prices. Transaction prices are instrumented using the exposure to the SDLT reform instrument. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table A9: Public library opening hours and housing prices – **OLS**, **TWFE**

Dependent variable	Hours nearest library		Hours within 1km	
	(1)	(2)	(3)	(4)
Panel A: Linear				
Housing value	1.0007***	0.9793***	1.6101***	1.5867***
	(0.3130)	(0.3104)	(0.3439)	(0.3415)
Dependent variable mean	37.892	37.892	40.036	40.036
R^2	0.58533	0.58538	0.75325	0.75333
Observations	605,870	605,870	182,423	182,423
Panel B: Adding square				
Housing value	1.6903	1.6714	-3.5423	-3.5137
	(3.9451)	(3.9439)	(4.1053)	(4.1060)
Housing value square	-0.0265	-0.0266	0.1969	0.1949
	(0.1507)	(0.1507)	(0.1574)	(0.1574)
Dependent variable mean	37.892	37.892	40.036	40.036
R^2	0.58533	0.58538	0.75330	0.75338
Observations	605,870	605,870	182,423	182,423
Turning point	31.854	31.384	8.9966	9.0148
Turning point decile	10	10	1	1
Regression specification:				
LSOA FE	X	X	X	X
Year FE	X	X	X	X
Controls		X		X

Notes: The table presents estimated coefficients of a regression of public library quality on neighborhood affluence, measured as (log) mean transaction prices. Transaction prices are instrumented using the exposure to the SDLT reform instrument. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table A10: Infrastructure investments and share of people with university degree – **OLS**

Dependent variable	Distance		# 1km		# Presence		Investment value	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Linear								
Share with degree	-0.6762	-1.6202**	0.5472*	0.5899*	0.0881***	0.0434	-0.1962	-0.2005
	(0.7154)	(0.8080)	(0.3307)	(0.3423)	(0.0340)	(0.0412)	(0.1380)	(0.1426)
Dependent variable mean	34.960	34.960	0.24755	0.24755	0.03615	0.03615	11.681	11.681
R^2	0.98310	0.98327	0.09083	0.09335	0.01213	0.01283	0.82518	0.82521
Observations	66,026	66,026	66,026	66,026	66,026	66,026	48,643	48,643
Panel B: Adding square								
Share with degree	0.9542	-2.8597	-0.3027	0.7036	0.1625	0.2022**	-0.4179	-0.4323
	(2.3263)	(2.4506)	(1.0518)	(1.1588)	(0.1042)	(0.1018)	(0.3229)	(0.3337)
Share with degree square	-2.3262	1.7557	1.2126	-0.1610	-0.1061	-0.2250	0.3262	0.3384
	(2.5733)	(2.6203)	(1.2532)	(1.3750)	(0.1433)	(0.1513)	(0.5948)	(0.6099)
Dependent variable mean	34.960	34.960	0.24755	0.24755	0.03615	0.03615	11.681	11.681
R^2	0.98310	0.98328	0.09085	0.09335	0.01214	0.01284	0.82520	0.82523
Observations	66,026	66,026	66,026	66,026	66,026	66,026	48,643	48,643
Regression specification:								
LAD x Year FE	X	X	X	X	X	X	X	X
Controls		X		X		X		X

Notes: The table presents estimated coefficients for the relationship between infrastructure investments and neighborhood affluence, measured as the share of people with a university degree. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table A11: Infrastructure investments and household income – **OLS**

Dependent variable	Distance		# 1km		# Presence		Investment value	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Linear								
Log net weekly income	0.5928	-0.4479	-0.8265**	-0.5955*	-0.1238*	-0.1183	0.1280**	0.1121**
	(0.4853)	(0.5199)	(0.3704)	(0.3354)	(0.0703)	(0.0759)	(0.0497)	(0.0507)
Dependent variable mean	23.165	23.165	0.16655	0.16655	0.02726	0.02726	11.253	11.253
R^2	0.98184	0.98204	0.02842	0.03153	0.00577	0.00621	0.76842	0.76860
Observations	164,595	164,595	164,595	164,595	164,595	164,595	104,254	104,254
Panel B: Adding square								
Log net weekly income	6.2586	-1.6089	-20.0916	-18.8313	-3.9542	-3.8979	-1.7650	-1.9991
	(10.9454)	(11.0933)	(16.2677)	(17.0111)	(3.4724)	(3.6072)	(1.4597)	(1.4488)
Log net weekly income square	-0.4421	0.0905	1.5031	1.4221	0.2989	0.2947	0.1487	0.1658
	(0.8283)	(0.8401)	(1.2426)	(1.3023)	(0.2658)	(0.2757)	(0.1162)	(0.1153)
Dependent variable mean	23.165	23.165	0.16655	0.16655	0.02726	0.02726	11.253	11.253
R^2	0.98184	0.98204	0.02901	0.03205	0.00586	0.00629	0.76845	0.76863
Observations	164,595	164,595	164,595	164,595	164,595	164,595	104,254	104,254
Regression specification:								
LAD x Year FE	X	X	X	X	X	X	X	X
Controls		X		X		X		X

Notes: The table presents estimated coefficients for the relationship between infrastructure investments access and neighborhood affluence, measured as (log) average estimated net weekly household income. The unit of observation is an aggregation of neighborhoods (MSOA). Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table A12: Public library quality and share of people with university degree – **OLS**

Dependent variable	Hours nearest library		Hours within 1km	
	(1)	(2)	(3)	(4)
Panel A: Linear				
Share with degree	1.5004	3.7162***	10.2784***	7.6638***
	(1.4217)	(1.3513)	(1.8184)	(1.6257)
Dependent variable mean	36.313	36.313	38.210	38.210
R^2	0.43138	0.45723	0.43665	0.50425
Observations	66,026	66,026	19,646	19,646
Panel B: Adding square				
Share with degree	-8.1274**	5.4174	4.0503	4.5838
	(4.0391)	(3.8057)	(5.3697)	(4.9189)
Share with degree square	13.7364**	-2.4096	8.3324	4.1142
	(5.9462)	(5.5811)	(7.6052)	(6.7075)
Dependent variable mean	36.313	36.313	38.210	38.210
R^2	0.43176	0.45724	0.43684	0.50429
Observations	66,026	66,026	19,646	19,646
Regression specification:				
LAD x Year FE	X	X	X	X
Controls		X		X

Notes: The table presents estimated coefficients for the relationship between public amenity access and neighborhood affluence, measured as the share of people with a university degree. The unit of observation is a neighborhood. Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

Table A13: Public library quality and household income – **OLS**

Dependent variable	Hours nearest library		Hours within 1km	
	(1)	(2)	(3)	(4)
Panel A: Linear				
Log net weekly income	-4.5267***	0.5175	-5.0395***	-0.7786
	(1.2166)	(1.1311)	(1.9146)	(1.5698)
Dependent variable mean	37.619	37.619	39.418	39.418
R^2	0.39438	0.42312	0.36306	0.42786
Observations	164,595	164,595	53,885	53,885
Panel B: Adding square				
Log net weekly income	-92.8344***	-37.6984	-166.2485***	-88.4547**
	(30.1354)	(28.6272)	(40.5630)	(37.7749)
Log net weekly income square	6.8900***	2.9802	12.5857***	6.8436**
	(2.3526)	(2.2501)	(3.1352)	(2.9446)
Dependent variable mean	37.619	37.619	39.418	39.418
R^2	0.39495	0.42322	0.36562	0.42861
Observations	164,595	164,595	53,885	53,885
Regression specification:				
LAD x Year FE	X	X	X	X
Controls		X		X

Notes: The table presents estimated coefficients for the relationship between public library access and neighborhood affluence, measured as (log) average estimated net weekly household income. The unit of observation is an aggregation of neighborhoods (MSOA). Controls include distance to CBD, distance to townhouses, an urban-rural dummy, and population density. Standard errors provided in parentheses are clustered at the district level with stars indicating *** p <0.01, ** p <0.05, and * p <0.1.

C Classification of tenders

```
                  tender_title
                        <char>
1: Birchwood Basketball Court

tender_description

<char>
1: This is a request for a quote to install a full size outdoor basketball court (without
fencing) into a grassed area.  Applications are by email to town.clerk@hatfield-herts.gov.
Appointments to view the site can be arranged through 01707 280401.
```

Figure A4: Tender classified as WORKS

D Amenity quality

In the main analysis, I have looked at the extensive margin effects of the presence of physical infrastructure in response to changes in neighborhood affluence. I measure amenity quality using two approaches: (1) opening hours of public libraries from the FOI requests, which proxies for facility size and accessibility; (2) infrastructure investment spending from procurement tenders, which measures maintenance and improvement spending. These measures capture different facets of quality: availability (hours) and investment (upgrades).

Tables A5 to A8 present the OLS and 2SLS regressions, which take the same form as before, just with new outcome variables. For the public libraries, the outcome variables are the opening hours of the nearest library and the cumulative opening hours of all libraries within 1 km. For the infrastructure investments, the dependent variables are the distance to the closest investment, the number of investments within 1 km, the number of investments in a neighborhood, and the investment value of the nearest investment. I again include local authority-specific time trends to isolate purely cross-sectional variation. As before, the same pattern seems to hold, namely that there is a U-relationship between the quality of public amenities and neighborhood affluence across neighborhoods within a local authority.

Results for amenity quality (Tables A5 - A8) show less consistent patterns than for amenity presence (Tables 2 - 5). For library opening hours, the cross-sectional analysis (OLS) suggests a negative relationship with house prices, while the 2SLS results are unstable, particularly for hours within 1km (Table A6, Columns 3–4). For infrastructure investments, the U-shaped relationship documented for amenity presence is partially replicated for investment counts but not for investment values.

The cross-sectional analysis above isolates variation within local authorities but at a single point in time. Panel structure allows an alternative approach: examining how amenities change within neighborhoods as they grow richer over time. I employ a two-way fixed effects model with neighborhood and year fixed effects (equations 1 and 2 below), which identifies the effect of price growth on amenity quality, controlling for neighborhood-specific time-invariant characteristics. This approach addresses concerns about time-invariant omitted variables but may be more susceptible to reverse causality (amenity improvements driving price increases). In contrast to the extensive margin analysis from above, where I look at big physical buildings like libraries, parks, or pools, infrastructure investments and opening hours are a margin that is easier to adjust and where we see more variation over time (see Appendix Figure A3 which shows the variation of infrastructure investments by type over time). Thus, I run the following regressions

$$y_{ilt} = \alpha_i + \tau_t + \beta_1 \log(\text{ price }_{ilt}) + \mathbf{X}'_{ilt}\theta + \psi_{ilt} \tag{47}$$

$$y_{ilt} = \alpha_i + \tau_t + \beta_1 \log(\text{ price }_{ilt}) + \beta_2 \log(\text{ price }_{ilt})^2 + \mathbf{X}'_{ilt}\theta + \iota_{ilt} \tag{48}$$

where the dependent variables are the same quality measures as before, but now α_i denotes

neighborhood fixed effects and τ_t denotes year fixed effects. Tables A9 and 6 present the results of these regressions and show a different picture. When looking at variation across neighborhoods within jurisdictions, a U-relationship between the availability and quality of public amenities is revealed. On the other hand, when looking at variation over time, neighborhoods that grow richer see more infrastructure investments and quality improvements of public amenities in their vicinity. Thus, while on the extensive margin, poor and rich areas have similar access to public amenities, quality improvements are done primarily in rich areas[12].

Cross-sectional and time-series analyses reveal different patterns. Across neighborhoods within local authorities (Panels B of Tables 2 - A8), the relationship between neighborhood affluence and amenity access is U-shaped: the poorest and richest neighborhoods have better access. However, within neighborhoods over time (Tables A9 - 6), as neighborhoods grow richer, they experience more infrastructure investments and improved library opening hours. This divergence suggests two distinct mechanisms: (1) Political economy factors may create a cross-sectional U-shape, where poor neighborhoods receive amenities as a form of redistribution or to address needs, and rich neighborhoods receive amenities through political power. (2) Over time, as neighborhoods gentrify, local political coalitions shift, and investment flows toward newly affluent areas. These mechanisms are explored in Section 5.

[12] Appendix Tables A10 – A13 show the same regressions but for different measures of neighborhood affluence and confirm this relationship

E Model Derivations

This appendix derives the main model objects used in Section 6. Residence locations are indexed by $r, i \in \{1, \ldots, S\}$, workplace locations by $m, j \in \{1, \ldots, S\}$, and worker types by $\theta \in \{L, H\}$. Workers choose a residence-workplace pair (r, m), consume the numeraire good C, rent housing floor space H, and value residential amenities B_r^θ. Workplace wages are denoted by w_m^θ, the local floor-space price by r_r, and bilateral commuting costs by d_{rm}.

E.1 Household Problem and Indirect Utility

The type-specific residential amenity index is

$$B_r^\theta = b_r g_r^{\eta_\theta} x_r^{\rho_\theta}, \tag{49}$$

where b_r is a fundamental residential amenity, g_r is the public amenity, and x_r is the private amenity. Conditional on choosing residence r and workplace m, worker ω has utility

$$U_{rm}^\theta(\omega) = \frac{z_{rm}(\omega) B_r^\theta}{d_{rm}} \left(\frac{C_{rm}^\theta(\omega)}{\beta} \right)^\beta \left(\frac{H_{rm}^\theta(\omega) - \bar{H}}{1-\beta} \right)^{1-\beta}, \qquad H_{rm}^\theta(\omega) \geq \bar{H}. \tag{50}$$

Disposable labor income is

$$I_m^\theta = (1-\tau) w_m^\theta. \tag{51}$$

The household budget constraint is

$$C_{rm}^\theta(\omega) + r_r H_{rm}^\theta(\omega) = I_m^\theta. \tag{52}$$

The subsistence housing requirement $\bar{H}$ does not enter as a separate expenditure item because households pay rent on total housing consumption $H_{rm}^\theta(\omega)$. It restricts feasible choices through $H_{rm}^\theta(\omega) \geq \bar{H}$, and it affects indirect utility through the cost $r_r \bar{H}$ of satisfying the housing floor.

Define housing above the subsistence level by

$$\widetilde{H}_{rm}^\theta(\omega) = H_{rm}^\theta(\omega) - \bar{H}. \tag{53}$$

The budget constraint can then be written as

$$C_{rm}^\theta(\omega) + r_r \widetilde{H}_{rm}^\theta(\omega) = I_m^\theta - r_r \bar{H}. \tag{54}$$

Thus, after paying for the housing floor, the household allocates remaining income $I_m^\theta - r_r \bar{H}$ between numeraire consumption and housing above the minimum. The problem has an interior solution only if

$$I_m^\theta - r_r \bar{H} > 0. \tag{55}$$

The first-order conditions imply the standard Cobb-Douglas expenditure shares over C and

$\widetilde{H}$:

$$C_{rm}^{\theta*} = \beta \left(I_m^\theta - r_r \bar{H} \right), \tag{56}$$

$$\widetilde{H}_{rm}^{\theta*} = \frac{(1-\beta)\left(I_m^\theta - r_r \bar{H} \right)}{r_r}. \tag{57}$$

Therefore optimal total housing demand is

$$H_{rm}^{\theta*} = \bar{H} + \frac{(1-\beta)\left(I_m^\theta - r_r \bar{H} \right)}{r_r} = \beta \bar{H} + \frac{(1-\beta) I_m^\theta}{r_r}. \tag{58}$$

Substituting the optimal choices into utility gives indirect utility

$$V_{rm}^\theta(\omega) = z_{rm}(\omega) v_{rm}^\theta, \tag{59}$$

where the deterministic component is

$$v_{rm}^\theta = \frac{b_r g_r^{\eta_\theta} x_r^{\rho_\theta}}{d_{rm}} \frac{\left[I_m^\theta - r_r \bar{H} \right]_+}{r_r^{1-\beta}}. \tag{60}$$

The operator $[\cdot]_+$ sets the term to zero when the housing floor is not affordable.

E.2 Fréchet Choice Probabilities

The idiosyncratic terms $z_{rm}(\omega)$ are independently Fréchet distributed with common shape parameter $\varepsilon > 1$:

$$F(z) = \Pr\left[z_{rm}(\omega) \leq z \right] = \exp\left(-z^{-\varepsilon} \right). \tag{61}$$

This is the common-scale version of the Fréchet structure used in Ahlfeldt et al. (2015). Equivalently, the scale parameters that shift average idiosyncratic utility across residence-workplace pairs are normalized to one. Thus, all systematic differences in the attractiveness of residence-workplace pairs are captured by the deterministic component v_{rm}^θ.

The indirect utility of type θ worker ω from residence-workplace pair (r, m) is

$$V_{rm}^\theta(\omega) = z_{rm}(\omega) v_{rm}^\theta. \tag{62}$$

For the derivation, first consider pairs with $v_{rm}^\theta > 0$. Pairs with $v_{rm}^\theta = 0$, which occur when the housing floor is not affordable, receive zero choice probability. Since V_{rm}^θ is a monotone

transformation of z_{rm}, the distribution of utility for pair (r, m) is

$$\begin{aligned} G_{rm}^{\theta}(u) &= \Pr\left[V_{rm}^{\theta}(\omega) \leq u\right] \\ &= \Pr\left[z_{rm}(\omega) \leq \frac{u}{v_{rm}^{\theta}}\right] \\ &= \exp\left[-\left(\frac{u}{v_{rm}^{\theta}}\right)^{-\varepsilon}\right] \\ &= \exp\left[-\left(v_{rm}^{\theta}\right)^{\varepsilon} u^{-\varepsilon}\right]. \end{aligned} \tag{63}$$

Define the pair-specific attractiveness term

$$\Phi_{rm}^{\theta} = \left(v_{rm}^{\theta}\right)^{\varepsilon}. \tag{64}$$

Using equation (60), this can be written as

$$\Phi_{rm}^{\theta} = \left(b_r g_r^{\eta_\theta} x_r^{\rho_\theta} \left[I_m^{\theta} - r_r \bar{H}\right]_+\right)^{\varepsilon} \left(d_{rm} r_r^{1-\beta}\right)^{-\varepsilon}. \tag{65}$$

With this notation, the utility distribution in equation (63) is

$$G_{rm}^{\theta}(u) = \exp\left[-\Phi_{rm}^{\theta} u^{-\varepsilon}\right]. \tag{66}$$

The corresponding density is

$$g_{rm}^{\theta}(u) = \frac{\partial G_{rm}^{\theta}(u)}{\partial u} = \varepsilon \Phi_{rm}^{\theta} u^{-(\varepsilon+1)} \exp\left[-\Phi_{rm}^{\theta} u^{-\varepsilon}\right]. \tag{67}$$

Workers choose the residence-workplace pair that delivers the highest indirect utility. Let

$$\Phi^{\theta} = \sum_{i=1}^{S} \sum_{j=1}^{S} \Phi_{ij}^{\theta}. \tag{68}$$

Because utility draws are independent across residence-workplace pairs, the CDF of the maximum utility draw is the product of the pair-specific CDFs:

$$\begin{aligned} G^{\theta}(u) &= \Pr\left[V_{ij}^{\theta}(\omega) \leq u \ \forall i, j\right] \\ &= \prod_{i=1}^{S} \prod_{j=1}^{S} G_{ij}^{\theta}(u) \\ &= \exp\left[-\left(\sum_{i=1}^{S} \sum_{j=1}^{S} \Phi_{ij}^{\theta}\right) u^{-\varepsilon}\right] \\ &= \exp\left[-\Phi^{\theta} u^{-\varepsilon}\right]. \end{aligned} \tag{69}$$

Thus the maximum of the pair-specific Fréchet utilities is itself Fréchet distributed, with scale summarized by Φ^{θ}.

The joint choice probability is the probability that pair (r, m) yields utility u, while every other pair yields utility weakly below u, integrated over all possible values of u:

$$\begin{aligned}\pi^{\theta}_{rm} &= \Pr\left[V^{\theta}_{rm}(\omega) \geq V^{\theta}_{ij}(\omega)\ \forall i, j\right] \\ &= \int_0^{\infty} \left[\prod_{(i,j)\neq(r,m)} G^{\theta}_{ij}(u)\right] g^{\theta}_{rm}(u)\, du. \end{aligned} \tag{70}$$

Substituting equations (66) and (67) gives

$$\begin{aligned}\pi^{\theta}_{rm} &= \int_0^{\infty} \exp\left[-\sum_{(i,j)\neq(r,m)} \Phi^{\theta}_{ij} u^{-\varepsilon}\right] \varepsilon \Phi^{\theta}_{rm} u^{-(\varepsilon+1)} \exp\left[-\Phi^{\theta}_{rm} u^{-\varepsilon}\right]\, du \\ &= \int_0^{\infty} \varepsilon \Phi^{\theta}_{rm} u^{-(\varepsilon+1)} \exp\left[-\Phi^{\theta} u^{-\varepsilon}\right]\, du. \end{aligned} \tag{71}$$

To evaluate this integral, note that

$$\frac{\partial}{\partial u} \exp\left[-\Phi^{\theta} u^{-\varepsilon}\right] = \varepsilon \Phi^{\theta} u^{-(\varepsilon+1)} \exp\left[-\Phi^{\theta} u^{-\varepsilon}\right]. \tag{72}$$

Therefore

$$\begin{aligned}\pi^{\theta}_{rm} &= \frac{\Phi^{\theta}_{rm}}{\Phi^{\theta}} \int_0^{\infty} \varepsilon \Phi^{\theta} u^{-(\varepsilon+1)} \exp\left[-\Phi^{\theta} u^{-\varepsilon}\right]\, du \\ &= \frac{\Phi^{\theta}_{rm}}{\Phi^{\theta}} \left[\exp\left[-\Phi^{\theta} u^{-\varepsilon}\right]\right]_{u=0}^{u=\infty} \\ &= \frac{\Phi^{\theta}_{rm}}{\Phi^{\theta}}. \end{aligned} \tag{73}$$

Since $u^{-\varepsilon} \to \infty$ as $u \to 0$ and $u^{-\varepsilon} \to 0$ as $u \to \infty$, the bracketed term equals $1 - 0$. Combining equations (65) and (73), the joint probability is therefore

$$\pi^{\theta}_{rm} = \frac{\Phi^{\theta}_{rm}}{\Phi^{\theta}}, \qquad \Phi^{\theta} = \sum_{i=1}^{S} \sum_{j=1}^{S} \Phi^{\theta}_{ij}. \tag{74}$$

Residence probabilities are obtained by summing joint probabilities over workplaces:

$$\pi^{\theta}_{Rr} = \sum_{m=1}^{S} \pi^{\theta}_{rm} = \frac{\sum_{m=1}^{S} \Phi^{\theta}_{rm}}{\Phi^{\theta}}. \tag{75}$$

Workplace probabilities are obtained by summing over residences:

$$\pi^{\theta}_{Mm} = \sum_{r=1}^{S} \pi^{\theta}_{rm} = \frac{\sum_{r=1}^{S} \Phi^{\theta}_{rm}}{\Phi^{\theta}}. \tag{76}$$

The probability of commuting to workplace m conditional on residence r is

$$\pi^{\theta}_{rm|r} = \frac{\pi^{\theta}_{rm}}{\sum_{j=1}^{S} \pi^{\theta}_{rj}} = \frac{\Phi^{\theta}_{rm}}{\sum_{j=1}^{S} \Phi^{\theta}_{rj}}. \tag{77}$$

For a fixed residence r, the residential amenity term $b_r g_r^{\eta_\theta} x_r^{\rho_\theta}$ and the floor-space price term $r_r^{1-\beta}$ are common across all workplace choices and cancel. The cost of the housing floor, $r_r \bar{H}$, does not disappear from the surplus-income term because it determines how much income a worker living in r has left when employed in each possible workplace. The conditional commuting probability is therefore

$$\pi^{\theta}_{rm|r} = \frac{\left(\frac{\left[I^{\theta}_m - r_r \bar{H}\right]_+}{d_{rm}}\right)^{\varepsilon}}{\sum_{j=1}^{S} \left(\frac{\left[I^{\theta}_j - r_r \bar{H}\right]_+}{d_{rj}}\right)^{\varepsilon}}. \tag{78}$$

The denominator is the type-specific commuting market access of residents of r:

$$CMA^{\theta}_r = \sum_{j=1}^{S} \left[I^{\theta}_j - r_r \bar{H}\right]_+^{\varepsilon} d_{rj}^{-\varepsilon}. \tag{79}$$

It summarizes the workplace opportunities available to type-θ residents of r: each workplace is weighted by after-tax earnings net of the cost of satisfying the housing floor in residence r, and discounted by commuting costs.

Similarly, the probability of living in residence r conditional on working in workplace m is

$$\pi^{\theta}_{rm|m} = \frac{\pi^{\theta}_{rm}}{\sum_{i=1}^{S} \pi^{\theta}_{im}} = \frac{\Phi^{\theta}_{rm}}{\sum_{i=1}^{S} \Phi^{\theta}_{im}}. \tag{80}$$

Substituting for Φ^{θ}_{rm} gives

$$\pi^{\theta}_{rm|m} = \frac{\left(b_r g_r^{\eta_\theta} x_r^{\rho_\theta} \left[I^{\theta}_m - r_r \bar{H}\right]_+\right)^{\varepsilon} \left(d_{rm} r_r^{1-\beta}\right)^{-\varepsilon}}{\sum_{i=1}^{S} \left(b_i g_i^{\eta_\theta} x_i^{\rho_\theta} \left[I^{\theta}_m - r_i \bar{H}\right]_+\right)^{\varepsilon} \left(d_{im} r_i^{1-\beta}\right)^{-\varepsilon}}. \tag{81}$$

For a fixed workplace m, the wage w^{θ}_m is common across all residence choices, but the surplus-income term $[I^{\theta}_m - r_i \bar{H}]_+$ still varies across residences because the cost of satisfying the housing floor depends on the residential floor-space price r_i. This is the key Stone-Geary difference relative to a Cobb-Douglas housing specification with $\bar{H} = 0$: if $\bar{H} = 0$, the wage term would be common across residences and would cancel from $\pi^{\theta}_{rm|m}$; with $\bar{H} > 0$, housing costs interact with workplace income in determining the attractiveness of each residence-workplace pair.

E.3 Population Aggregates

Let N^{θ} denote the total population of type θ. Residence populations are

$$H^{\theta}_{Rr} = N^{\theta} \pi^{\theta}_{Rr} = N^{\theta} \sum_{m=1}^{S} \pi^{\theta}_{rm}. \tag{82}$$

Workplace employment by type is

$$H^{\theta}_{Mm} = \sum_{r=1}^{S} \pi^{\theta}_{rm|r} H^{\theta}_{Rr}. \tag{83}$$

Total residence and workplace populations are

$$H_{Rr} = \sum_{\theta} H^{\theta}_{Rr}, \qquad H_{Mm} = \sum_{\theta} H^{\theta}_{Mm}. \tag{84}$$

The high- and low-type residence shares are

$$s_{rH} = \frac{H^{H}_{Rr}}{H_{Rr}}, \qquad s_{rL} = \frac{H^{L}_{Rr}}{H_{Rr}} = 1 - s_{rH}. \tag{85}$$

E.4 Local Welfare and the Council's First-Order Condition

Average utility of type θ residents in neighborhood r is the residence-population-weighted average across workplace choices:

$$\bar{U}^{\theta}_{r}(g_r) = \frac{1}{H^{\theta}_{Rr}} \sum_{m=1}^{S} H^{\theta}_{Rrm} U^{\theta}_{rm}(g_r), \qquad H^{\theta}_{Rrm} = \pi^{\theta}_{rm|r} H^{\theta}_{Rr}. \tag{86}$$

For the local welfare aggregator, the model uses the Atkinson/CES form

$$W_r(g_r) = \begin{cases} \left[s_{rL} \left(\bar{U}^{L}_{r}(g_r) \right)^{1-\chi} + s_{rH} \left(\bar{U}^{H}_{r}(g_r) \right)^{1-\chi} \right]^{\frac{1}{1-\chi}}, & \chi \neq 1, \\ \left(\bar{U}^{L}_{r}(g_r) \right)^{s_{rL}} \left(\bar{U}^{H}_{r}(g_r) \right)^{s_{rH}}, & \chi = 1. \end{cases} \tag{87}$$

The main specification sets $\chi = 1$, so local welfare is the weighted geometric mean of type-specific utilities. Since public amenities enter type-specific utility as $g_r^{\eta_\theta}$, type-specific average utility is proportional to $g_r^{\eta_\theta}$, holding the residence composition fixed when evaluating the government's static allocation problem. For $\chi = 1$,

$$\log W_r(g_r) = s_{rL} \log \bar{U}^{L}_{r}(g_r) + s_{rH} \log \bar{U}^{H}_{r}(g_r). \tag{88}$$

Differentiating gives

$$\frac{\partial \log W_r(g_r)}{\partial g_r} = \frac{s_{rL}\eta_L + s_{rH}\eta_H}{g_r}. \tag{89}$$

Therefore

$$\frac{\partial W_r(g_r)}{\partial g_r} = W_r(g_r) \frac{\bar{\eta}_r}{g_r}, \qquad \bar{\eta}_r = s_{rL}\eta_L + s_{rH}\eta_H. \tag{90}$$

Aggregate welfare is

$$W(g) = \sum_{r=1}^{S} \frac{H_{Rr}}{N} W_r(g_r), \qquad N = \sum_{r=1}^{S} H_{Rr}. \tag{91}$$

The political component of public amenity provision is

$$P(g) = \sum_{r=1}^{S} p(s_{rH}) g_r^{\psi}, \qquad p(s_{rH}) = \alpha_L s_{rL}^2 + \alpha_H s_{rH}^2. \tag{92}$$

The council maximizes

$$U^p(g) = (1-\gamma)P(g) + \gamma W(g) \tag{93}$$

subject to the balanced-budget constraint

$$\sum_{r=1}^{S} \xi g_r = T. \tag{94}$$

Tax revenue is

$$T = \tau \sum_{\theta} \sum_{m=1}^{S} w_m^{\theta} H_{Mm}^{\theta}. \tag{95}$$

The Lagrangian for the council problem is

$$\mathcal{L}^p = (1-\gamma) \sum_{r=1}^{S} p(s_{rH}) g_r^{\psi} + \gamma W(g) + \lambda \left(T - \sum_{r=1}^{S} \xi g_r \right). \tag{96}$$

For an interior allocation, the first-order condition for each neighborhood r is

$$(1-\gamma)\psi p(s_{rH}) g_r^{\psi-1} + \gamma \frac{\partial W(g)}{\partial g_r} = \lambda \xi. \tag{97}$$

Using separability of aggregate welfare and equation (90),

$$\frac{\partial W(g)}{\partial g_r} = \frac{H_{Rr}}{N} W_r(g_r) \frac{\bar{\eta}_r}{g_r}. \tag{98}$$

The council's first-order condition can therefore be written as

$$(1-\gamma)\psi p(s_{rH}) g_r^{\psi-1} + \gamma \frac{H_{Rr}}{N} W_r(g_r) \frac{\bar{\eta}_r}{g_r} = \lambda \xi. \tag{99}$$

E.5 Production and Commercial Floor-Space Demand

In the focal quantitative exercise, wages are taken as an exogenous type-location schedule. Production is used to map workplace employment and wages into commercial floor-space demand. Firms in workplace m produce the numeraire good using labor and commercial floor space:

$$Y_m = A_m H_{Mm}^{\alpha} L_{Cm}^{1-\alpha}. \tag{100}$$

With perfect competition and Cobb-Douglas production, the wage bill equals the labor share of output:

$$WB_m = \sum_{\theta} w_m^{\theta} H_{Mm}^{\theta} = \alpha Y_m. \tag{101}$$

The remaining share of output is paid to commercial floor space:

$$r_m L_{Cm} = (1-\alpha) Y_m. \tag{102}$$

Combining these two zero-profit conditions gives commercial floor-space demand:

$$L_{Cm} = \frac{1-\alpha}{\alpha} \frac{WB_m}{r_m}. \tag{103}$$

E.6 Residential Floor-Space Demand and Rent Clearing

Residential floor-space demand follows from the household housing demand in equation (58). Let

$$\bar{y}_r^\theta = \sum_{m=1}^{S} \pi_{rm|r}^\theta I_m^\theta \tag{104}$$

denote average disposable income for type θ residents of neighborhood r. Aggregating individual housing demand over worker types and workplace choices gives

$$L_{Rr} = \sum_\theta \sum_{m=1}^{S} \pi_{rm|r}^\theta H_{Rr}^\theta \left(\beta \bar{H} + \frac{(1-\beta) I_m^\theta}{r_r} \right) \tag{105}$$

$$= \beta \sum_\theta H_{Rr}^\theta \bar{H} + \frac{1-\beta}{r_r} \sum_\theta H_{Rr}^\theta \bar{y}_r^\theta. \tag{106}$$

The first term is the floor space required to satisfy the subsistence housing requirement. The second term captures additional housing demand above this minimum. Conditional on satisfying the housing floor, households allocate a share $1-\beta$ of remaining disposable income to housing, so this component rises with average disposable income and falls with the local floor-space price.

Total floor-space supply in each location i is fixed at L_i and is allocated between residential and commercial uses:

$$L_i = L_{Ri} + L_{Ci}. \tag{107}$$

Using equations (103) and (106), market clearing in location i requires

$$L_i = \beta \sum_\theta H_{Ri}^\theta \bar{H} + \frac{1-\beta}{r_i} \sum_\theta H_{Ri}^\theta \bar{y}_i^\theta + \frac{1-\alpha}{\alpha} \frac{WB_i}{r_i}. \tag{108}$$

Solving for the equilibrium floor-space price gives

$$r_i = \frac{(1-\beta) \sum_\theta H_{Ri}^\theta \bar{y}_i^\theta + \frac{1-\alpha}{\alpha} WB_i}{L_i - \beta \sum_\theta H_{Ri}^\theta \bar{H}}. \tag{109}$$

When $\bar{H} = 0$, the residential component collapses to the Cobb-Douglas housing-demand case.

E.7 Equilibrium Conditions

Given fundamentals $\{b_r, x_r, L_r\}_{r=1}^S$, commuting costs $\{d_{rm}\}_{r,m=1}^S$, exogenous type-location wages $\{w_m^\theta\}_{m=1,\theta\in\{L,H\}}^S$, total type populations $\{N^\theta\}_{\theta\in\{L,H\}}$, and model parameters, an equilibrium is a collection

$$\left\{g_r, r_r, \pi_{rm}^\theta, H_{Rr}^\theta, H_{Mm}^\theta, L_{Rr}, L_{Cr}\right\}_{r,m,\theta} \tag{110}$$

such that: household choice probabilities satisfy equation (73); residence populations satisfy equation (82); workplace employment satisfies equation (83); commercial floor-space demand satisfies equation (103); rents clear floor-space markets according to equation (109); public amenities satisfy the council first-order condition in equation (99); and the public-amenity budget clears:

$$\sum_{r=1}^{S} \xi g_r = T. \tag{111}$$